Drivers of rate dispersion in loan markets: information acquisition or search frictions?

VITTORIA IANNOTTA*

June 23, 2026

Click here for the most updated version.

ABSTRACT

Using French loan-level data, I document substantial dispersion in interest rates among otherwise similar firms, even when they borrow from the same bank. I further show that unexplained rate residuals correlate with surprises in firms' future credit risk outcomes, in line with an information channel in loan pricing. The central question is the extent to which observed dispersion reflects true differences in borrower risk versus the extra-value that banks manage to extract. To address this, I develop and calibrate a model of loan pricing with imperfect competition and costly bank screening. Dispersion can be the result of costly banks' screening, where the acquisition of negative signals on firms translates into higher rate offers; or the optimal bank strategy in a non-competitive environment, in which a deviation from the law of one price is possible and banks can extract firms' surplus. The calibrated results suggest that imperfect competition is the prevailing force driving observed dispersion.

*HEC Paris. This work has benefited from a State grant managed by the Agence Nationale de la Recherche under the Investissements d'Avenir programme with the reference ANR-18-EURE-0005 / EUR DATA EFM.

1 Introduction

Interest rates in credit markets exhibit substantial *interest rate fragmentation across firms and banks*. A substantial part of this dispersion is accounted for by macroeconomic conditions and fundamental factors, including bank-firm relationships, contract characteristics and observable credit risk - up to $80 - 85\%$ for the Euro Area (Altavilla et al. (2024)). However, an empirical regularity is the persistence of a significant unexplained component, a residual dispersion that survives after controlling for these observed fundamentals. This residual, as shown in a growing literature on rate dispersion and in this paper, should not be treated as noise, but rather reflects structural forces of the banking loan market.

One perspective is to focus on rate dispersion *across similar firms and loan products*. Amiti et al. (2026) use US large loan data to document that dispersion persists within bins of homogeneous firms and loan characteristics, originated in the same quarter. They entirely attribute this across-firms dispersion to search frictions. Similarly, Kozeniauskas (2024) argues that "idiosyncratic noise" arises from differing search abilities, preventing firms from securing the lowest available rates. In addition, in both papers the contribution of firms' default risk seems negligible, as the share of variance explained by default risk measures is fairly low. This finding is in contrast with a large literature on adverse selection, that suggests that the role of unobservables and banks' relationship lending matters for interest rates. For instance, Claessens et al. (2025) use internal bank ratings assigned by banks to show that private information component matters in determining lending rates.

More puzzling, using French loan-level data, we show that dispersion may persist not only across similar firms and products, but also within the same bank (*across firms, within bank*). To rationalize this finding, we reconcile the literature on dispersion and on private information by jointly considering search frictions and unobserved risk as determinants of the unexplained rate dispersion. Considering both drivers and understanding the nature of unexplained rate dispersion is crucial, as they have opposite implications on pricing efficiency and policy intervention. On the one hand, dispersion arising in an imperfectly competitive environment is generating mispricing across firms, with important consequences on firms' investment decisions and firm competition. In addition, monetary policy pass-through is impaired by banks' market power, and overall this provides a rationale for regulation and policy intervention. On the other hand, dispersion induced unobserved information on firms' true credit risk reflects the efficiency of the banking market in overcoming asymmetric information. Any policy intervention is at best unnecessary, and policy makers can be reassured about the full pass-through of monetary policy to rates. Moreover, the interaction between these forces has non-trivial implications for the transmission of monetary policy, as we will show in the paper. After showing evidence that both forces are at play, we will propose a framework to study their interaction and understand which is prevailing.

Firstly, leveraging detailed micro-data on lending in France, spanning from 2006 to 2024, we characterize the time and cross-sectional distribution of unexplained rate dispersion. Time-bank fixed effects and observable firm and loan characteristics account for 84% of total rate variation, which is consistent with the findings in Altavilla et al. (2024) for Euro area loans. The distribution of unexplained rate residuals over time shows significant dispersion and time variation, and it responds to monetary policy shocks - a thightening reduces total variance. Secondly, we show evidence that unexplained dispersion is not pure noise, but is related to measures of competition and unobserved information. Firms in more concentrated markets face higher interest rates, but also lower overall dispersion. This suggests that banks manage to squeeze rate offers towards higher rates. Moreover, we show that rates contain forward information about the future evolution of firms' creditworthiness. In particular, there is a positive and significant correlation between paying a relatively higher rate and being ex-post more likely to default. We exclude that this result is driven by a leverage effect - leverage increases, fundamentals deteriorate, leading to higher default probability. In fact, this effect is not stronger for larger loans. Also, we test the transmission of interest rate changes to firm-level risk, exploiting the interest rate structure of the contract, fixed versus floating. Consequently, these residuals contain banks' acquired information on firms' riskiness, orthogonal to observed characteristics, which feeds into the loan interest rate. Moreover, we show that in contexts where information matters the most, this correlation is stronger; using age as a proxy of the firms' track record in credit markets, we show that for younger firms this correlation is significantly larger than for relatively older firms.

We move to studying a tractable theoretical framework that determines loan pricing as the outcome of imperfect competition, modeled via search frictions, and asymmetric information with information acquisition by banks about observationally equivalent firms. The model features two symmetric banks and two firm types, differing in their risk (success probability); search frictions are modelled as in Burdett and Judd (1983). In addition, banks can acquire a symmetric signal on firms. We characterize the equilibrium dispersed rate distribution, showing that both frictions jointly contribute to observed dispersion. Search frictions deliver *within-type dispersion*. In a non-perfectly competitive environment, banks compete for borrowers, but search frictions prevent them from perfectly undercutting each other; consequently, even homogeneous face different rates as a mixed-strategy equilibrium arises. The information structure of the market affects banks' incentives to pool/separate borrowers, determining *between-type dispersion*; under perfect information, or when banks acquire a very precise signal, good types will pay on average lower rates than bad types, leading to a positive correlation between rates and default probability.

Second, we derive important implications on the welfare distribution. Under search frictions only, but perfect information, all firms match and maximum welfare is achieved; the stronger the search frictions, the larger the banks' share of total surplus. More interestingly, the information structure has dramatic implications on welfare losses and cross-subsidization. When adverse selection is severe, good types are either cross-subsidizing bad types in a pooling equilibrium, or are fully pushed out of the market, with consequent welfare losses; more information increases overall welfare and reduces the subsidy.

Finally, we derive implications on monetary policy transmission. As expected, limited competition generally weakens the pass-through of policy rates; in fact, as rates are already close to the maximum rate that firms are willing to accept, an increase in policy rates will only partially pass through to the rate distribution. Moreover, we show that this effect interacts with the information structure of the market. Via the cost of funds, monetary policy affects the total gains from trade for each type, but mostly for safer firms; consequently, under information asymmetry, higher rates move the equilibrium from a pooling to a separating one, with safe firms either heavily subsidizing riskier ones or being completely pushed out of the market. We derive and validate the model's predictions regarding monetary policy transmission. The model predicts that, following an increase in the policy rate, the funding cost for banks is higher, and total observed dispersion decreases. Using high-frequency monetary policy shocks, we find a reduction in the dispersion of interest rate residuals.

To get a sense of the relative magnitudes, we run a calibration exercise to recover the implied parameters of competition and signal precision from the unexplained rate distribution across Banque de France rating classes. We target 3 data moments: the total variance of rate residuals, decreasing in competition and increasing in signal precision; the correlation between rates and riskiness, increasing in both competition and signal precision; the maximum rate observed. The exercise suggests that the equilibrium is close to perfect information in ex-ante safer markets, with information decreasing as we move to ex-ante riskier rating classes. However, the relative contribution of competition to unexplained variance is predominant (0-5%), even under perfect information. This result hides substantial differences in surplus distribution across banks and types.

The remainder of the paper is organized as follows. Section 2 documents the literature. Section 3 and 4 presents the data and stylized facts. Section 5 outlines the theoretical framework. Section 6 details the estimation and quantitative results. Section 7 investigates the monetary policy implications.

2 Related Literature

Our paper connects several strands of literature: the empirics of interest rate dispersion, the theory of search frictions and market power in credit markets, the literature on information acquisition and screening in banking, and the study of monetary policy transmission through the credit channel.

Rate dispersion in loan markets. In credit markets, rate dispersion after controlling for observable fundamentals is the object of many papers leveraging loan-level data in different contexts. Cavalcanti et al. (2021) document dispersion across observationally equivalent firms in Brazil and focus on the consequences on capital misallocation and economic development; they show how this mechanism can be a driver of divergence in economic development across country. Altavilla et al. (2024) decompose the variation in rates on Euro Area loans into macroeconomic fundamentals, bank identity, firm and loan characteristics, which overall account for 80–85%

of total rate variation. Our measure of rate variation is consistent to theirs, but we will focus on the unexplained component. Using French data, Mazet-Sonilhac (2025) provides evidence on the existence of search frictions in the French loan market, which motivates a theory of bank-firm matching where search and information frictions interact. We engage in the ambitious exercise of disentangling the relative contribution of both frictions to observed dispersion. Amiti et al. (2026) use US large-loan data to document dispersion within narrow bins of observationally equivalent firms and quarters; measures of banks' private information on default risk (internal ratings) fail in explaining rate variation, and thus they entirely attribute it to search frictions. In our setting, lacking direct measures of banks' private information, we will find an innovative way to extract the information component: we use the future evolution of firms' credit risk, as summarized by external credit ratings. Kozeniauskas (2024) similarly documents "idiosyncratic noise" in financing rates arising from heterogeneous search abilities, with default risk explaining little of the residual variation. In line with these papers, we find that search frictions play a predominant role in explaining rate dispersion, calling for policy intervention. However, we depart from these papers by showing that the low contribution of banks' private information on default risk on rate dispersion does not imply that banks are not screening at all. On the contrary, we will find that banks have strong screening incentives, in particular in buckets of firms with ex-ante low risk; omitting this component can mislead the understanding of the role of screening in loan pricing, and upwardly biases estimates of the contribution of search frictions to observed dispersion.

Interactions of search frictions and screening in credit markets. From a theoretical perspective, *price dispersion as an equilibrium outcome* has deep theoretical roots. Varian (1980) shows that when consumers differ in their search behavior, identical sellers optimally randomize over prices, generating dispersion without any quality differences. Asymmetric information in credit markets has equally deep roots: from Akerlof (1970)'s insight that information asymmetry can unravel markets, to Stiglitz and Weiss (1981) who show it leads to credit rationing even in competitive loan markets. Our theoretical framework builds on Burdett and Judd (1983) - who show that when a fraction of buyers compare multiple prices, banks randomize over rates in a dispersed price equilibrium without risk differences - and more closely to Lester et al. (2019) who introduce asymmetric information and screening via menus in a market of buyers and sellers facing search frictions. We do not find evidence of a positive relation between rates and quantities in our setting, consequently we model screening via costly information acquisition in a credit market with search and asymmetric information. We find overall similar implications on welfare implications and cross-subsidization from good to bad firms. *Disentangling search frictions and information* is challenging as they most likely interact in our setting. Banks' private information is a source of market power, and search frictions have consequences on the birth of new lending relationships and bank's screening ability. Carletti et al. (2024) provide a comprehensive review of how market power and informational advantages interact in banking. The interaction between competition and information acquisition has been directly theorized in credit markets: Hauswald and Marquez (2006) show that banks invest in information technology to screen borrowers, but stronger competition reduces the return on this investment and leads to under-screening; Dell'Ariccia and Marquez (2006) show that lending booms driven by intensified competition generate looser lending standards, as banks cannot recoup screening costs when rivals can poach identified good borrowers. Both papers provide theoretical grounding for the competition-information trade-off at the heart of our model. On the empirical side, Claessens et al. (2025) use internal bank ratings to isolate the private information component in loan terms, finding that it significantly affects rates, collateral, and maturity; Agarwal and Hauswald (2010) show that geographic proximity to the borrower reduces informational asymmetries and improves pricing accuracy. Central to our interpretation is the *hold-up problem*: Rajan (1992)[1] formalizes how a bank that acquires private information gains a competitive advantage, allowing it to extract surplus even from high-quality borrowers; Cahn et al. (2024) show that the public disclosure of credit ratings attenuates this hold-up by reducing banks' informational monopoly. Ioannidou and Ongena (2010) document that firms switching banks receive rate discounts, consistent with incumbents extracting rents from informationally captive borrowers. Flanagan (2023) estimates that screening costs are embedded in loan rates and are larger for more opaque borrowers, directly linking information acquisition to pricing. Related sources of market power include *switching costs and relationship rents*: Drechsler et al. (2017) demonstrate that concentrated deposit markets allow banks to maintain wide spreads; Petersen and Rajan (1995) show that greater competition reduces relationship lending;[2] Degryse and Ongena (2005) document spatial price discrimination, with banks charging higher rates to distant borrowers who are less likely to search; and Boot and Thakor (2000) show that intensified competition erodes banks' incentives to invest in relationship-based screening. Scharfstein and Sunderam (2016) provide direct evidence that market power in mortgage origination dampens monetary policy pass-through. The paper most closely related to ours is Crawford et al. (2018), who jointly estimate asymmetric information and market power in Italian mortgages and find both frictions quantitatively important; we differ by making information acquisition endogenous and by deriving explicit monetary policy implications. In a mortgage market setting, Agarwal et al. (2024) develop a structural model combining search frictions with lender screening, finding that both jointly determine credit allocation and calling for extensions to the business loan market, which this paper pursues. In our setting, we assume for simplicity that search frictions are exogenous to market conditions; however, bank's screening incentives will depend on the degree of competition in the market via their effect on expected profits.

Search frictions and information in other financial markets. The search friction framework has been productively applied *beyond bank credit*. Duffie et al. (2005) develop the foundational search model for over-the-counter (OTC) financial markets, showing how search costs generate bid-ask spreads and dealer rents. Farboodi et al. (2025) show that intermediaries extract rents precisely because search frictions give them market power over their clients. In the interbank money market, Bech and Monnet (2015) apply a random search model to explain the dispersion of overnight rates and trading volumes in OTC reserve markets. On the theoretical side, Guerrieri

[1] Building on Sharpe (1990), who first formalizes how banks exploit relationship information to extract rents from borrowers even in a competitive market.

[2] For a survey of relationship lending, see Boot (2000); Berger and Udell (1995) and Berger and Udell (2002) provide foundational empirical evidence on how relationship banking benefits small firms.

et al. (2010) characterize equilibrium in competitive search markets with adverse selection: high-quality sellers post higher prices that attract longer queues, generating a positive price-quality correlation in equilibrium. This mechanism is closely related to the rate-risk correlation we document in our data, and to the separating equilibria that arise in our model under information acquisition.

Monetary policy transmission. The transmission of monetary policy through the credit market is shaped by both market structure and information frictions. Bernanke and Gertler (1995) establish the credit channel: policy tightening raises external finance premia, amplifying real effects for bank-dependent borrowers. Holmström and Tirole (1997) provide the micro-foundations: firms with sufficient net worth access monitored bank credit at lower premia, while capital-constrained firms are rationed, so tightening disproportionately affects the latter. Kashyap and Stein (2000) identify the bank lending channel empirically using cross-sectional variation in bank liquidity: less liquid banks reduce loan supply more sharply during monetary contractions. Jiménez et al. (2012) provide loan-level evidence that monetary contractions reduce credit supply, particularly for weaker banks and borrowers; Jiménez et al. (2014) show that low rates encourage excessive risk-taking by banks. More related to our setting, Puglisi (2023) documents dispersion in the US auto loan market and shows that retail lending rate dispersion generates asymmetric monetary policy pass-through. On the market structure side, Drechsler et al. (2017) document that deposit market power dampens pass-through, and Wang et al. (2022) find that bank market power attenuates the transmission to corporate investment. Delis et al. (2025) show that tighter credit conditions fall disproportionately on lower-income borrowers, consistent with information frictions amplifying distributional effects. We identify monetary policy shocks using the high-frequency event-study approach of Altavilla et al. (2019), who construct the benchmark Euro Area monetary policy database. We contribute to this debate by showing how the interplay of search frictions and information acquisition shapes the rate distribution: pooling equilibria induced by mild adverse selection create rigidities that further impair the transmission of policy shocks.[3]

3 Data

This paper draws mainly on three firm-level datasets provided by the Banque de France. It leverages in particular detailed loan issuance data at the loan level, which provide all loan terms and relevant firm characteristics which help to predict interest rates. This information is complemented with credit registry data and balance-sheet information on firms.

1. Nouveaux Crédits (2006-2024) *Source: Banque de France*

The *Nouveaux Crédits* is a quarterly survey in which banks report the universe of newly originated loan and credit lines contracts to businesses in the first month of each quarter. It includes all loan terms (rate, amount, maturity, loan purpose and type, fixed or variable rate, guarantees), for both new agreements and renegotiated contracts that specify interest rates for the first time or revise key financial terms. This dataset is matched to

[3]More broadly, search frictions in credit markets amplify aggregate shocks and generate slow recoveries; see Den Haan et al. (2003) and Wasmer and Weil (2004).

firm-level identifiers (anonymized SIREN) and enriched with information on firms' descriptives, accounting and credit information (main accounting variables like balance-sheet size, debt structure, short term due repayments, proportion of drawn and undrawn credit lines, revenues). It includes also the credit score assigned by Banque de France (outlined in the following section). This dataset allows us to estimate the expected interest rate by firm-loan category, offering insights into the pricing of credit, and particularly useful for studying the dispersion of interest rates, the role of firm risk, and the transmission of monetary policy. While we focus on term loans, we store information on additional products issued in the same month (bundles).

2. Credit Registry (Service Central des Risques, SCR) (2006-2025) *Source: Banque de France*

The *Service Central des Risques* (SCR) is a credit registry operated by the Banque de France, which collects monthly data on outstanding loans granted by credit institutions and investment firms to companies. The reporting threshold is set at €25,000 per lender-borrower pair, which is high enough to study bank relationships for medium and large firms, excluding micro entreprises. The SCR serves as a supervisory risk database, tracking exposures across the financial system. The dataset will be particularly useful to control for borrowing patterns, credit constraints, and long-term firm-bank relationships.

3. FIBEN (1989-2022) *Source: Banque de France*

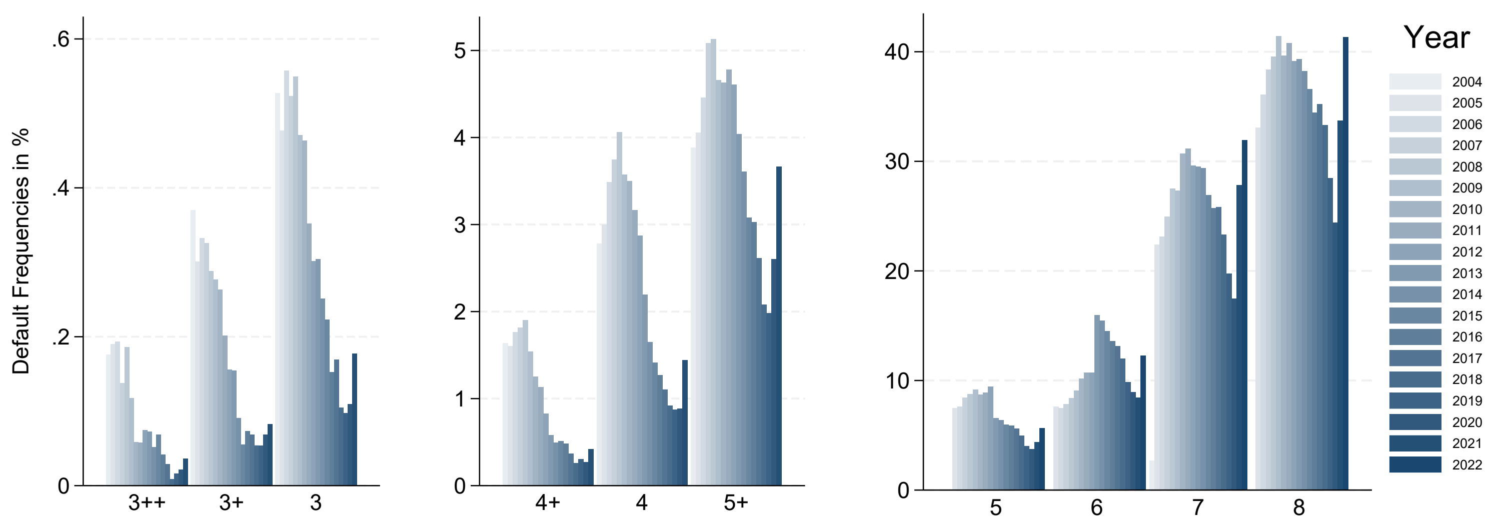
The *Fichier Bancaire des Entreprises* (FIBEN) is the Banque de France's firm-level accounting database. It draws original data from the tax returns filed by French companies, and includes aggregates and ratios that are relevant for financial assessment purposes. It includes financial statements (balance sheet, income statement, and off-balance sheet items), which are submitted annually to the tax authorities. The included accounting information will be very useful to assess firms' financial health and computing ex-post performance of firms. While the Banque de France does not collect all forms or all variables, the database still provides rich firm-level financial data over long time periods.

4. COTA (1989-2025) *Source: Banque de France*

This dataset contains the external rating provided by Banque de France to companies, with details on dates of revisions and relevant credit events. This rating is assigned by Banque de France annually on the basis of the latest balance-sheets and credit data available to firms beyond a sales threshold of 750K, increased in 2024 to 1250K. It is a forward-looking measure, as it gives an assessment of the firm's creditworthiness over a three-year period. Banque de France explicitly states to use the following information: the latest balance-sheet data available; data on default/delay events in payments (CIPE); sector-specific information; qualitative information, including interviews with firm managers on an annual basis, to assess the risks and difficulties the firm is facing. The rating entry is made of two component, summarizing firm size and creditworthiness; we will focus on the second one. As shown in Fig. A.1, the rating scale includes 12 rating categories, including two entries for liquidation (P) and a (0) when there is not enough information to assign a rating. A higher rating means higher default probability. This scale was reformed to a more granular one from January 2022; we will use the old scale and use Banque de France's conversion table for new categories.

From rating data to default frequencies.
Following Banque de France's approach to rating evaluations, we compute historical 3-year ahead default frequencies by rating class and year. For example, in year x we follow all firms with a rating equal to y on 01/01/x and compute the share of firms who defaulted on 31/12/x+2. The resulting default frequencies are reported in Fig. 1. Since we are interested in the relative distance between types and transition to other rating classes, we fix default frequencies to one year (2021) and use it throughout the sample as representing the probability of default by rating class. In practice, this means that we only care about firms who change rating class over time, and we disregard the changes of PD frequencies within one class from one year to the other. We check for robustness that our analysis goes through also using default frequencies by year, which is explained by the fact that the default frequencies by class maintain the same order of magnitude over time.

Figure 1: **3-year ahead default frequencies, by rating class and year (2004-2022).**



4 Stylized facts

4.1 Interest Rate Dispersion

Using loan-evel data, we first measure interest rate dispersion, within banks and between firms, after controlling for observable loan and firm characteristics.

We start from all credit issued in the first month of each quarter, from 2006q1 to 2022q4. We focus on fixed-term loan products to private companies, and exclude credit lines/other types of credit with undefined maturity. We are left with 1.9 million observations and 0.9 million unique firms respectively. We report a descriptive table of main loan and firm characteristics for the regression sample, by rating class. The average loan is around 1 milion, with substantial heterogeneity between rating classes, and it represents between one fourth and one third of the total firm's bank debt.

Rating Scale	**N obs.**	**rate** (mean)	**rate** (s.d.)	**Quantity** (mean, K-Eur)	**Maturity** (mean, months)	**Var. Rate** (in %)	**Age** (mean, years)	**Loan/Assets** (mean, in %)	**Bank Debt/Assets** (mean, in %)
3++	7948	2.1	1.8	3754.5	59.0	35	39.0	5.4	19.6
3+	22681	2.2	1.8	2919.3	53.5	37	33.7	5.0	20.6
3	43073	2.3	1.8	2271.5	42.4	37	31.7	5.3	22.0
4+	71629	2.8	2.1	1092.2	34.8	39	28.1	6.2	23.0
4	112974	2.8	2.0	670.2	27.9	43	26.1	7.1	25.0
5+	106418	2.9	2.0	545.5	23.9	48	23.4	7.7	30.3
5	46510	3.5	2.2	781.8	17.9	51	23.1	8.3	31.0
6	13625	3.7	2.2	923.4	16.6	52	21.4	10.4	34.2
7	2657	4.2	2.3	134.7	14.8	62	19.7	8.6	32.7
8	1592	5.0	2.6	107.8	13.2	61	18.0	8.4	31.2
9	268	5.5	2.7	134.0	11.9	48	16.5	10.0	31.7
P (liquidation)	3716	5.0	2.0	114.5	3.4	26	21.8	5.7	33.4
0 (no rating)	24777	3.2	2.1	1895.8	45.0	40	16.1	12.4	27.2
Total	457868	2.9	2.1	1098.1	30.5	44	25.8	7.3	26.5

Table 1: Source: Banque de France (Nouveaux Credits, COTA, SCR, FIBEN) Descriptive table of the loan-level regression sample. Rate refers to total effective rate (including all fees).

Baseline Approach. We follow an approach that is vary similar to the one by Altavilla, Gürkaynak, Quaedvlieg (2024)[4], by gradually inclding fixed effects from the macro- (month, lender) to micro- (loan and firm characteristics) level, and then including loan and firm characteristics. For each firm-bank-loan-time observation $\text{Rate}_{i,b,l,t}$, we run the following regression:

$$\text{Rate}_{i,b,l,t} = \alpha + \beta \cdot \text{LoanChar}_{i,b,l,t} + \gamma \cdot \text{FirmChar}_{i,t} + \delta_{FE} + \varepsilon_{i,b,l,t} \tag{1}$$

where time t is month, b is bank, i is firm, l is loan; X includes loan size, risk weighting assigned by the bank, maturity, variable rate dummy, personal loan dummy, individual entrepreneur dummy, firm age and age squared, risk weighting, short term debts, turnover, collateral (% of assets, if available), available credit before limit, bank and non bank debt (potentially also past volatility from balancesheet data). Fixed effects include Month × Banks × Loan type × 2-digit Sector, which well captures the supply schedule for similar products for each bank; then we include region, size, rating, size, fixed effects, and whether the firm belongs to a group. The regression sample includes 458 thousand observations and 120 thousand firms.

[4]Altavilla, Carlo and Gürkaynak, Refet S. and Quaedvlieg, Rogier, Macro and Micro of External Finance Premium and Monetary Policy Transmission (April, 2024). ECB Working Paper No. 2024/2934

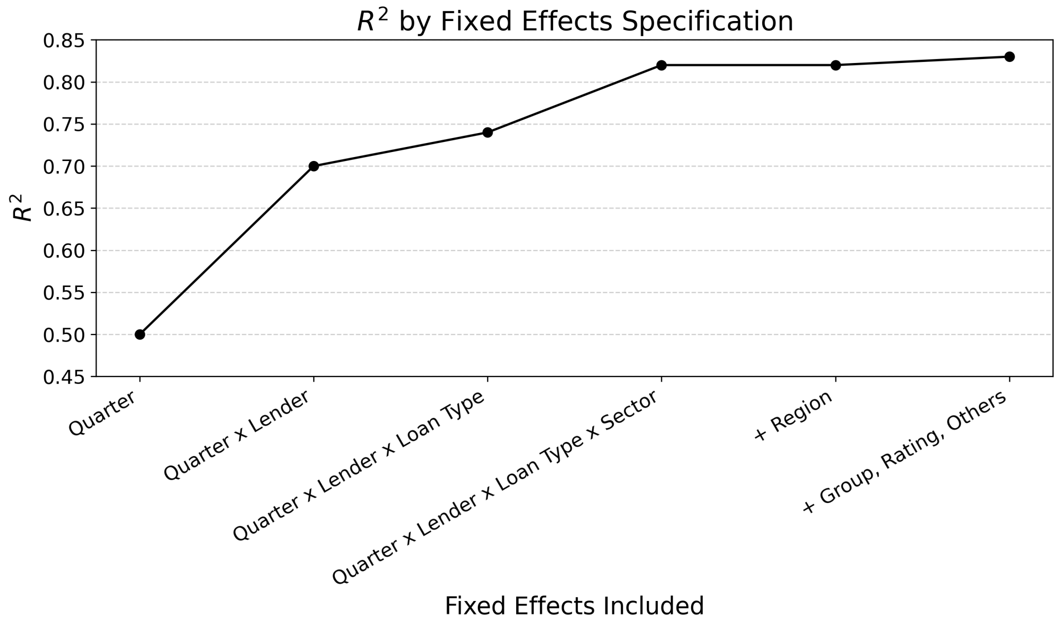


Figure 2: The figure shows the increase in R^2 of the baseline regression after gradually including fixed effects interactions and the other observables. Observables explain up to 84% of total rates variation.

Fig. 2 summarizes the share of variation explained by observables, while the regression table is reported in appendix Table B.1. The explained variation on observables is around 84%, with most of the variation explained by month (50%) and lender (20%) interaction, and the remaining 10% explained by loan and firm level characteristics.

For robusteness, we run alternative specifications: we add interaction terms (month-bank-french department) to the specification above, without sibstantial improvements in explained variation; or other regression models, to account for the fact that by contruction there are no rates below zero (Tobit, truncated regression), but with worse explanatory performance. Finally, in Appendix B we report additional specifications, including a specification with firm fixed effects, varying dispersion by bank and the alternative regression approaches.

4.1.1 Time and cross-sectional distribution of rate residuals

The residuals of the above regression represent the unexplained rate component of the interest rate, and importantly is a measure of rate dispersion across firms, after accounting for the bank supply schedule. Rate dispersion is substantial, 1pp on average over the time period - a meaningful gap in the context of loan pricing and external financing in general - and substantial time variation over the two decades (see Appendix B).

4.2 Rate residuals and banks' screening

We propose a novel test to elicit the informational component in interest rates. We show that rate residuals are positively correlated with unpredicted firms' realized riskiness, as implied by the evolution of firms' rating assigned by Banque de France. This approach has several strengths. Using a measure of creditworthiness over balance-sheet or growth indicators better captures default expectations, as utimately the bank cares about loan repayment. In addition, while other papers measuring dispersion use the internal banks' assessment to explain interest rates, they do not usually check whether this measure effectively predicts future default. Finally, it solves the issue of few observations when using only actual defaults, as firms' rating is reviewed annually. After documenting that the correlation is positive and significant, we will exclude that it is purely driven by a leverage effect via a quasi-experimental design, by comparing the effect of a raise in rates between firms borowing at fixed an variable rate, and conclude that it reflects banks' screening.

4.2.1 Correlating rate residuals and changes in default probability

Rate residuals might be the result of two factors: i) *imperfect competition*, which endow banks with some degree of market power, which implies a deviation from competitive prices ii) *banks' expectations* about firms' true default risk, if banks acquire additional information with respect to what we observe. In both cases, although for different reasons, rates will be correlated with ex-post firm risk outcomes. If the bank manages to charge a higher interest rate to a firm, their leverage would increase with respect to other firms and their ex-post fundamentals will worsen, leading to a deterioration of the rating; on the contrary, if the rate should reflect beliefs about the true firm type, we should observe that banks correctly anticipate a rating deterioration. Importantly, from the interest rate regression on observable it does not emerge a positive correlation between loan quantity and rates, so we exclude screening via menus and rather focus on *active screening by banks, via information acquisition.*
We test for this correlation by including the changes probability of default of firm i P_i^{t+k} in our baseline rate regression, as a proxy of the true firm type at time t.

$$\text{Rate}_{i,b,l,t} = \alpha + \beta \cdot \text{LoanChar}_{i,b,l,t} + \gamma \cdot \text{FirmChar}_{i,t} + \sum_{t+1}^{t+k} \beta_k \Delta P_i^{t+k} + \delta_b + \lambda_t + \varepsilon_{i,b,l,t} \tag{2}$$

where time t is quarter, b is bank, i is firm, l is loan. Standard errors are clustered at the lender-quarter level. An increase in P_i^{t+k} captures the transition to a worse rating class after k years from loan issuance. Since we control for information at time t, the coefficients β_k measure the correlation with unexplained changes in the borrower's default risk. We plot the β_k over the horizon $[t+1, t+h]$ in Fig. 3. We find a positive and significant correlation between the rate residuals and the future change in default probability, which is stronger in the first year after the loan issuance and then decreases over time. This suggests that rates contain forward-looking information about the borrower's creditworthiness that is not captured by observable characteristics at the time of loan issuance. The fact that the correlation is stronger in the first year may indicate that banks have better information about short-term risks or that the impact of unobserved factors diminishes over time.

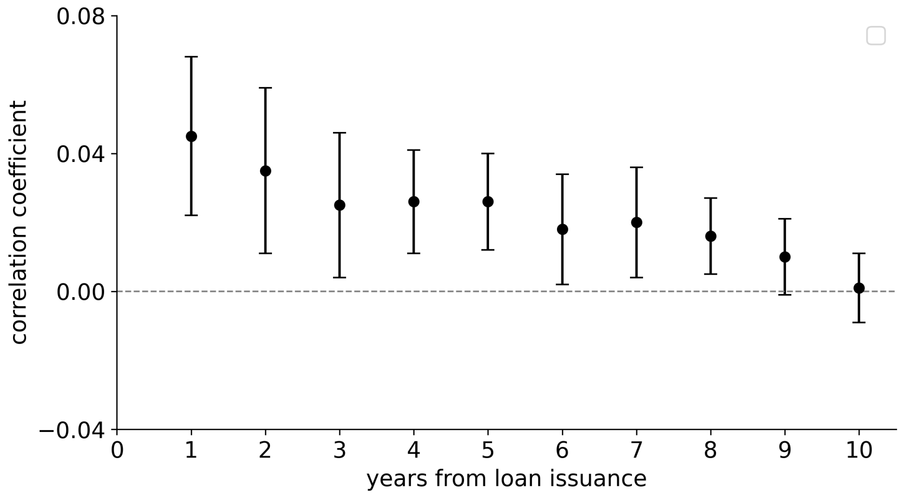


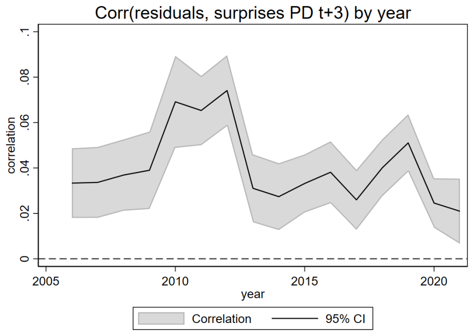
Figure 3: Correlation between rate residual and unexpected changes in default probability, by year. Balanced sample.

Excluding the leverage effect. To show that the observed correlation between rate residuals and unexpected changes in riskiness is not driven by a leverage effect - higher rates induce higher ex-post riskiness through the worsening of firm's fundamentals - we directly test the transmission of interest rate changes to firm-level risk in a quasi experimental setting. We exploit the different rate structure offered to firms (floating vs fixed rate) to test the impact of a rise in reference rates on firms' riskiness. We find no significant impact of increased debt-servicing costs on subsequent rating worsening, suggesting that an increase in servicing costs inducing an increase in leverage cannot alone induce a worsening firm-level risk. Interestingly, while the rate increase does not induce a worse ex-post outcome, the loan interest rate remains a significant predictor of ex-post riskiness. As the intensity of the effect depends on the individual exposure to variable rate, we further add an interaction term of the reference rate increase with the loan-to-assets and total bank debt-to-assets ratios; the result is unchanged. To further mitigate the issue of self-selection of firms into fixed or floating rate contracts, we repeat the exercise on loans originated at the end of the ZLB period, when firms did not expect a rate hike (end 2021), with same results. A more comprehensive treatment of this analysis is provided in the Appendix B.6.

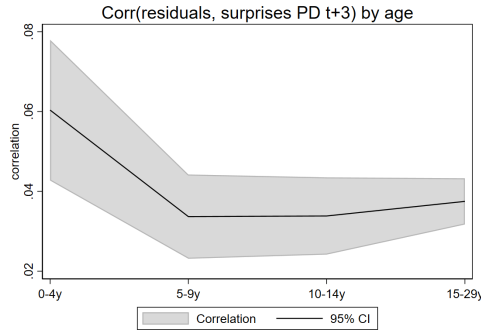
4.2.2 Banks' screening across firms' demographics

We show how the positive correlation between rate residuals and future credit deterioration varies across borrowers' demographics in Fig. 4. We first separate firms by their Banque de France **rating** at loan issuance into three broad categories. We document that the positive correlation is driven by firms with a relatively good rating, while it is not significant for relatively worse firms. This rejects to what a leverage effect would imply, as firms with higher default probabilities are closer to the default threshold and should be more sensitive to higher rates. Then, we introduce interaction terms between changes in default probabilities and i) **loan/bank debt share of assets**, ii) **age** in the above regression. This allows us to estimate and compare the correlation for significantly different quantiles of loan size and age, to further support the relevance of the information channel. While we expect the leverage effect to be stronger if the firm is paying a rate premium over a large proportion of its liabilities, we find that the correlation coefficients when comparing the 10th vs 90th or 20th vs 80th quantiles of loans over assets are not signicantly different. We repeat the same exercise with the share of bank debt, in case the residual was informative of the rates on other loans, and results are robust. Finally, we find that the correlation for younger firms (below the 20% percentile of age in our sample) is significantly larger than for older ones (above 80% percentile). This is in line with the idea that banks may put more effort in acquiring additional information or screening applications exactly when information is more valuable, as age is both a proxy for risk/growth prospects (Haltiwanger, Jarmin,Miranda, 2013) and also a proxy of the length in the credit market (Cloyne, Ferreira, Froemel, Surico, 2023). See Appendix B.6.

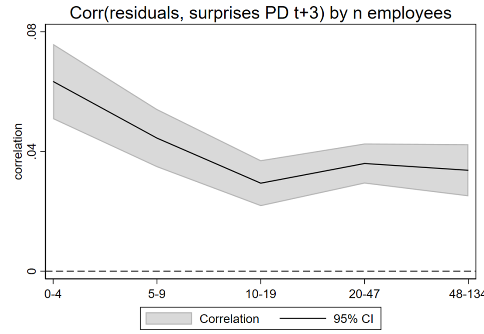
We further explore the strength of the correlation over time and in the cross-section. The above methods require many observations, so we rely on a different approach: we compute separately the rate residuals and the surprises in rating changes for the entire population, then we group firms over observables and extract the pure correlation between these terms. Three patterns emerge consistent with a private-information interpretation. First, the correlation varies over **time** (panel a), peaking around 2009–2012 at the height of the global financial crisis, when heightened uncertainty likely raised the return to screening, and declining during the zero lower bound period when default risk was broadly compressed. Second, the informational content of rates declines with **firm age** and size, as measured by **number of employees**: the correlation for firms aged 0–4 years (≈ 0.06) is roughly twice that for firms aged 10 years or more, and similarly higher for firms with fewer than 10 employees relative to larger ones (panel b,c). Younger and smaller firms have shorter credit histories and less public information available, leaving more room for bank-acquired private information to influence pricing. Using **asset size**, the correlation is relatively smaller for very lare firms, it becomes substantially larger for firms just below the 1 million threshold, and becomes non significant for very small firms, mainly due to lack of observations (panel d). Third, the correlation rises with **loan maturity** up to the 25–84 months range before plateauing: longer-horizon lending creates stronger incentives for banks to screen borrowers, since the value of acquiring information over the life of the contract is larger (panel e). Finally, we do not find clear patterns when grouping firms by the **length of the bank-firm relationship** (panel f).
All together, these cross-sectional patterns indicate that the informational content of unexplained rate dispersion is stronger where there is more adverse selection.



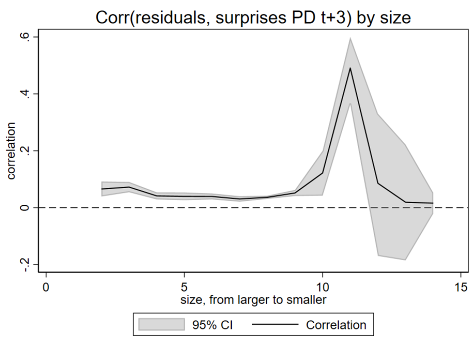
(a) By year



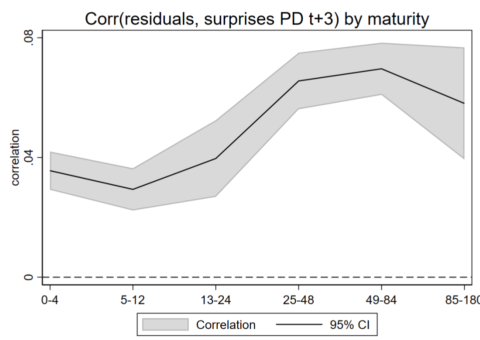
(b) By firm age



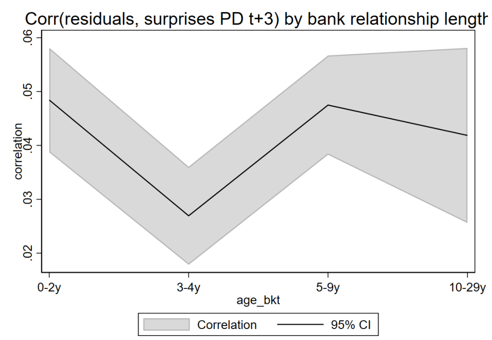
(c) By number of employees



(d) By firm size (revenue class)



(e) By loan maturity (months)



(f) By bank relationship length

Figure 4: *Informational content of rate residuals across borrower and loan characteristics.* Each panel shows the correlation $\hat{\rho}_g$ between the unexplained rate component and the unexpected 3-year change in default probability (ΔPD_{t+3}), both computed in the entire population and correlated by group g. Both variables are first residualized on the full set of observables from the baseline regression (firm financials, loan characteristics, quarter $\times$ bank $\times$ loan type $\times$ sector fixed effects), then correlated within groups. Shaded bands are 95% confidence intervals based on the Fisher z-transformation applied to $\hat{\rho}_g$. Panel (d) reports Banque de France revenue-based size categories ordered from largest firms (left) to smallest (right); wide confidence intervals for the smallest categories reflect limited observations.

5 Model

In the previous section we showed that interest rate dispersion, defined within banks and across firms, is informative of the degree of competition and screening in the banking sector. To quantify their relative role in determing observed dispersion, we introduce a model of loan pricing in which imperfect competition and asymmetric information are jointly at play. This model will be able to reproduce observed dispersion via imperfect compeition (*within-type dispersion*) and screening (*between-type dispersion*), and the positive correlation between rates and risk when banks recognise riskier types and charge higher rates. In the next section, we will calibrate this framework to provide an estimate of the relative importance of screening and search in our framework, and then test the monetary policy implications of the model.

The framework draws from the general market setting proposed by Lester et al. (2019), where competition is modeled via search frictions, and we adapt it to the bank loan market. However, we deviate from the assumption of menu contracts, for which we find little evidence in our sample[5]. Instead, we assume that banks can choose to acquire costly signals on firms true credit risk and adjust the interest rates, fixing quantity. For generality, we also introduce an interest rate "leverage effect".

5.1 Environment

We examine the interactions between two homogeneous and risk-neutral banks and a continuum of firms. Firm sample banks' offers to take one loan of fixed quantity 1 at an interest rate r to undertake a project. Projects are risky, characterized by a success probability p, which for generalization could be endogenous to the interest rate on the loan r - a higher interest rate linearly decreases the ex-post success probability:

$$p = p(r), \quad \text{with} \quad \frac{\partial p}{\partial r} < 0. \tag{3}$$

Consequently, the bank's effective cost of lending, denoted by c, becomes a function of the interest rate charged. As the rate rises, the default risk increases, implying:

$$c_i = c_i(r), \quad \text{with} \quad \frac{\partial c}{\partial r} > 0. \tag{4}$$

5.1.1 Gains from trade.

Banks. The opportunity cost of bank capital is set by the central bank's policy rate, r^{CB}. We assume *limited liability*, so firms only repay the loan if the project is successful. Therefore, the competitive break-even rate for banks, c, is defined by the condition that the expected repayment equals the opportunity cost of capital $c = \frac{r^{CB}}{p(c)}$.

Firms. There is a continuum of risk-neutral firms of measure 1 per bank. We assume that firms have a reservation rate R, which for simplicity represents the value of the project if successful[6], and that if they reject the rate offer they do not undertake the project and get a infitely small, strictly positive, value $\varepsilon > 0$. In other words, firms will only accept a loan if the interest rate is below their reservation rate. We also assume that firms are affected by monetary policy rates exclusively through bank lending.

5.1.2 Market Frictions

The model departs from the standard competitive framework by introducing two distinct types of frictions: imperfect competition via search frictions and asymmetric information regarding firm types.

Search Frictions. We introduce search frictions as in the dispersed equilibrium framework of Burdett and Judd (1983) (BJ), which generates imperfect competition between banks and rate dispersion not related to credit

[5] we find that, conditional on all observable characteristics in our sample, there is a negative relationship between rates and quantity
[6] it may also represent the minimum rate available in the non-bank market

risk. Specifically, we assume that a fraction π of firms are "shoppers", meaning they sample offers from both banks and accept the lowest one, while the remaining fraction $1-\pi$ are "captive", meaning they only sample one offer and accept it if it is below their reservation rate. Banks know the aggregate proportion π, but the specific number of offers a firm has sampled is private information to the firm. For expositional convenience, we assume that π is constant across firm types, although this assumption can be relaxed. This structure leads to a mixed-strategy equilibrium in interest rates.

Asymmetric Information. The population of firms is divided into two types, *Good* (share μ_g) and *Bad* (share μ_b), where $\mu_b+\mu_g=1$, differing in riskiness, as the probability of success is lower for bad firms: $p_b(r)<p_g(r)$. Consequently, for banks the cost of lending to a bad firm is strictly higher than lending to a good firm at a given interest rate r: $c_g(r)<c_b(r)$. In the following, we also assume that the reservation rate of good firms is lower than that of bad firms: $\bar{r}^g<\bar{r}^b$. This assumption is not crucial to solve the model, but it will be important to deliver the positive correlation between the firm's type and the interest rate paid. The bank cannot directly observe the type of a firm, but it knows the population shares and the type-specific cost functions and reservation rates. This structure leads to a separating or partially pooling equilibrium in interest rates, depending on the severity of the adverse selection problem. Finally, we look at the case in which $c_g(r)<c_b(r)<\bar{r}^g<\bar{r}^b$.

5.1.3 Information Acquisition

Let $\theta\in\{g,b\}$ denote the firm's true type, with population shares μ_g and $\mu_b=1-\mu_g$. Each bank can acquire a binary signal $s\in\{g,b\}$ at cost C_s per firm. The signal is informative but imperfect:

$$\Pr(s=\theta\mid\theta)=q_\theta>0.5,\quad\forall\theta\in\{g,b\}$$

where q_g is the probability of correctly identifying a good type, and q_b is the probability of correctly identifying a bad type.

The fraction of firms receiving each signal is:

$$\sigma_g\equiv\Pr(s=g)=q_g\mu_g+(1-q_b)\mu_b$$
$$\sigma_b\equiv\Pr(s=b)=q_b\mu_b+(1-q_g)\mu_g$$

By construction, $\sigma_g+\sigma_b=1$.

Bayesian updating. Upon observing signal $s_i\in\{g,b\}$, the bank updates its prior belief μ_g to a posterior belief about facing a good type:

$$\alpha^{s_g} \equiv \Pr(\theta = g \mid s = g) = \frac{q_g \mu_g}{q_g \mu_g + (1 - q_b)\mu_b}$$

$$\alpha^{s_b} \equiv \Pr(\theta = g \mid s = b) = \frac{(1 - q_g)\mu_g}{(1 - q_g)\mu_g + q_b \mu_b}$$

Note that $\alpha^{s_g} > \mu_g > \alpha^{s_b}$: a good signal increases the belief that the firm is good, while a bad signal decreases it. The complementary probabilities of facing a bad type are:

$$1 - \alpha^{s_g} = \frac{(1 - q_b)\mu_b}{q_g \mu_g + (1 - q_b)\mu_b}, \quad 1 - \alpha^{s_b} = \frac{q_b \mu_b}{(1 - q_g)\mu_g + q_b \mu_b}$$

5.2 Equilibrium Analysis

We start by examining the equilibrium properties of the model in the extreme cases of perfect information (or perfect signal) or no information (or no signal). The equilibrium will feature dispersed rate distributions: *within types* for any value of $\pi \in (0, 1)$, excluding extreme cases of monopoly and perfect competition; *between types*, depending on the underlying hypothesis on the information structure.

5.2.1 Equilibrium under perfect information (or perfect signal)

The bank perfectly recognizes the type when it meets a firm. Basically, this is equivalent to solving two type-specific maximization problems.

Profit maximization. Banks' expected profits are as follows:

$$Profits^i(r) = \begin{cases} \mu_i[(1 - \pi) + \pi(1 - F(r))](r - c_i) & \text{if } r \leq \bar{r} \\ 0 & \text{otherwise} \end{cases}$$

That is, the bank trades-off higher profits and lower acceptance probability: every firm i is met with probability μ_i; the bank gets $(r - c_i)$ if the firm accepts; the firm accepts with probability 1 if it is captive, and if it observes a higher offer from the other bank when it is a shopper, with probability $(1 - F(r))$. $F(r)$ is the equilibrium rate distribution.

In limit cases of monopoly ($\pi = 0$) or perfect competition ($\pi = 1$), the distribution is degenerate. The bank offers to each type the reservation rate $\bar{r}^i$ under monopoly, as all firms will always sample only one offer, and the competitive rate c_i under perfect competition, as all firms will visit both banks and get two rate quotes. However, under imperfect competition $\pi \in (0, 1)$, a dispersed rate equilibrium arises, as in BJ. Intuitively, any rate between the cost and the reservation rate could not be the pure-strategy equilibrium, as each bank would be tempted to slightly lower the offered rate and win all the shoppers, until they reach the competition rate; furthermore, at the competition rate, each bank would be making zero profits, and could make strictly positive ones by increasing the interest rate and make profit from captive firms. The type-specific equilibrium distribution can be found exactly like in BJ:

$$\mu_i(1-\pi)(\bar{r}^i - c_i) = \mu_i[1-\pi+\pi(1-F^i(r))](r-c_i)$$

(where we are assuming that, although the cost will be convex in r, the slope of c(r) at $\bar{r}^i$ is lower than 1).

Proposition 5.1. *For $\pi \in (0,1)$ and under perfect information (or perfect signal), there exists a unique symmetric mixed-strategy equilibrium. For each firm type i, banks randomize their rate offers according to the cumulative distribution function $F^i(r)$ on the support $[\underline{r}^i, \bar{r}^i]$:*

$$F^i(r) = \begin{cases} 0 & \textit{if } r < \underline{r}^i \\ 1 - \dfrac{1-\pi}{\pi}\left[\dfrac{\bar{r}^i - r}{r - c_i(r)}\right] & \textit{if } \underline{r}^i \leq r \leq \bar{r}^i \\ 1 & \textit{if } r > \bar{r}^i \end{cases}$$

where the upper bound $\bar{r}^i$ is the reservation rate, and the lower bound $\underline{r}^i$ is implicitly defined by $F(\underline{r_i}) = 0$.

Finally, the observed rate distribution across types is the mixture of the distributions, weighted by each type's population share.

$$F^i(r) = \begin{cases} 0, & \text{if } r < \underline{r^g} \\ \mu_g F^g(r) + \mu_b F^b(r), & \text{if } \underline{r^g} < r \leq \bar{r}^b \\ 1, & \text{if } r > \bar{r}^b \end{cases}$$

Densities f(r).:

$$f^i(r) = \frac{\partial F^i(r)}{\partial r} \qquad\qquad f(r) = \mu_g f^g(r) + \mu_b f^b(r)$$

Given a specific functional form for the cost function, we can solve for the dispersion (simply the variance of the total distribution) and the correlation between observed rates and default probabilities to match the empirical findings. We will solve and simulate a version of the model in the next section.

5.2.2 Equilibrium under no information (or uninformative signal)

The bank operates under information asymmetry, namely it cannot distinguish the firm type upon meeting. We may still assume that the bank knows the population composition and the type-specific costs and reservation rates. Consequently, it faces a type-specific demand on two separate intervals: if $r < \bar{r}^g$, both types would accept an offer (pooling), while if $\bar{r}^g < r < \bar{r}^b$ the bank knows that only bad types would accept. The equilibrium will feature a pooling or separating region depending on the severity of the adverse selection problem; mild adverse selection would incentivize pooling, while in the severe adverse selection case banks are more likely to reduce or avoid pooling and offer only high rates.

Limit cases of monopoly/competition $\pi = 0, 1$. We begin by analyzing the equilibrium in the extreme cases of monopoly and perfect competition to build the intuition for the general case.

Monopoly $\pi = 0$. The bank makes profits

$$\Pi = \begin{cases} (r - c^{avg}(r)) & \text{if } \underline{r} < r \leq \bar{r}^g \\ \mu_b(r - c_b(r)) & \text{if } \bar{r}^g < r \leq \bar{r}^b \end{cases}$$

where $c^{avg}(r) = \mu_g c_g(r) + \mu_b c_b(r)$ is the weighted average of the type-specific costs. The bank compares the maximum profit it can extract in each region:

$$\max\{(\bar{r^g} - c_{avg}(r^g)), \mu_b(\bar{r^b} - c_b(\bar{r^b}))\}$$

We assume that marginal costs increase sufficiently slowly ($c'(r) < 1$) such that profits are increasing in rates within each interval[7]. Consequently, the optimal strategy must be a corner solution: the bank compares the maximum profit from pooling (at $\bar{r}^g$) against the maximum profit from screening (at $\bar{r}^b$).

Lemma 5.1. ***Define ϕ:***

$$\boxed{\phi \equiv (\bar{r}^g - c^{avg}(\bar{r}^g)) - \mu_b(\bar{r}^b - c_b(\bar{r}^b))}$$

Under monopoly ($\pi = 0$), $c'(r) < 1$ and with an uninformative signal, the unique equilibrium is $\bar{r^g}$ if $\phi > 0$, and $\bar{r^b}$ if $\phi \leq 0$.

Under monopoly, the unique profit-maximizing interest rate depends on the sign of ϕ. **We can interpret ϕ and an indicator of the strength of asymmetric information.** If $\phi \leq 0$, adverse selection is severe, and banks extract more profits from trading with bad types only. If $\phi > 0$ adverse selection is mild, and banks extract more profits from pooling good and bad types.

[7] if $c'(r)$ was increasing over the rate interval, there may be a point where the cost is higher than the benefit, and an internal solution would be optimal ($r* < \bar{r^g}$ or $r* < \bar{r^b}$)

Perfect competition $\pi = 1$ In this case, the bank makes zero profits, offering the competitive rate in the population.

Lemma 5.2. *Under perfect competition ($\pi = 1$) and asymmetric information, the unique equilibrium is a pooling contract at the competitive rate $r^c = c^{avg}$, implicitly defined by the break-even condition given average costs. Given the assumption $c^b < \bar{r}^g$, this rate is strictly below the reservation rate of both types and is accepted by the entire population.*

When $\pi = 1$, firms always compare offers. Bertrand competition drives the interest rate down to the break-even point. Since the competitive rate satisfies $r^c < \bar{r}^g$ by assumption, both types participate, and the bank faces the average cost of the population. The equilibrium profit is therefore $\Pi = (\mu_b + \mu_g)(r^c - c^{avg}(r^c)) = 0$.

General case: Imperfect Competition. Bank profits depend on the interval.

$$\Pi = \begin{cases} [1-\pi+\pi(1-F(r))](r - c^{avg}(r)) & \text{if } r \leq \bar{r}^g \\ \mu_b[1-\pi+\pi(1-F(r))](r - c_b(r)) & \text{if } \bar{r}^g < r \leq \bar{r}^b \end{cases}$$

where $c^{avg}(r) = \mu_g c_g(r) + \mu_b c_b(r)$. As in the perfect information case, the bank trades-off profits and acceptance probability. However, the bank knows that both firms would accept a rate in the pooling region, while a rate on the separating region would be accepted by bad types and rejected by good type, that would exit the market. As in the perfect information case with one type, the equilibrium CDF will be derived as the isoprofit function given profits at the maximum rate charged. However, now the maximum rate depends on the severity of adverse selection, as captured by the value of ϕ.

Severe adverse selection: $\phi \leq 0$

Banks surely trade with bad types, and possibly offer some rates in the pooling region. When trading with captive firms, the bank makes more profits from trading with bad types than pooling. Therefore, the maximum rate observed in the market is $\bar{r}^b$. The bank offers a distribution of rates on the separating region to extract profits from bad types, and may also offer some rates in the pooling region to extract profits from good types, such that:

$$\begin{aligned} &[1-\pi+\pi(1-F^a(r))](r - c^{avg}(r)) = \mu_b(1-\pi)(\bar{r}^b - c_b(\bar{r^b})) && \text{if } r \leq \bar{r}^g \\ &\mu_b[1-\pi+\pi(1-F^a(\bar{r^g}) - F^b(r))](r - c_b(r)) = \mu_b(1-\pi)(\bar{r}^b - c_b(\bar{r^b})) && \text{if } \bar{r}^g < r \leq \bar{r}^b \end{aligned}$$

Proposition 5.2. *For $\pi \in (0,1)$, $\phi \leq 0$, $c'(r) < 1$ and with an uninformative signal, there exists a unique symmetric mixed-strategy equilibrium. Banks randomize their rate offers according to the cumulative distribution* *function $F(r)$ on the support $[\underline{r}^a, \bar{r}^b]$:*

$$F(r) = \begin{cases} 0 & \textit{if } r < \underline{r}^a \\ F^a(r) & \textit{if } \underline{r}^a \leq r \leq \bar{r}^g \\ F^a(\bar{r}^g) + F^b(r) & \textit{if } \bar{r}^g < r \leq \bar{r}^b \\ 1 & \textit{if } r > \bar{r}^b \end{cases}$$

where the equal profit condition on each interval gives $F^a(r), F^b(r)$:

$$F^a(r) = \min\left\{1 - \frac{1-\pi}{\pi}\left[\frac{\mu_b(\bar{r}^b - c_b(\bar{r}^b))}{(r - c^{avg}(r))} - 1\right], 0\right\}$$
$$F^b(r) = 1 - F^a(\bar{r^g}) - \frac{1-\pi}{\pi}\left[\frac{(\bar{r}^b - c_b(\bar{r}^b))}{(r - c_b(r))} - 1\right]$$

and the lower bounds $\underline{r}^a$ and $\underline{r}^b$ are implicitly defined by $F^a(\underline{r}^a) = 0$ and $F^b(\underline{r}^b) = 0$.

$$\underline{r}^a = c^{avg}(\underline{r}^a) + \mu_b(1-\pi)(\bar{r}^b - c_b(\bar{r}^b))$$
$$\underline{r}^b = c_b(\underline{r}^b) + \frac{\bar{r}^g - c^{avg}(\bar{r}^g)}{\mu_b}$$

This form ensures that the combined function on the two intervals cumulates to 1: by construction, $F^a(\bar{r^g}) + F^b(\bar{r^b}) = 1$. Our equilibrium differs from Burdett and Judd (1983) in accounting for demand heterogeneity, preserving the mixed-strategy equilibrium result. On the existence of a pooling region, we find that it depends on the severity of adverse selection, as captured by ϕ.

Lemma 5.3. *The pooling region exists when adverse selection is relatively mild $-\frac{\pi}{1-\pi}(\bar{r}^g - c^{avg}) < \phi \leq 0$, while for values of ϕ below this threshold the equilibrium is fully separating.*

We find the threshold from $F^a(\bar{r}^g) > 0$. Intuitively, when adverse selection is severe, the bank prefers to offer only separating contracts, as the profits from pooling are too low. When adverse selection is mild, the bank offers some pooling contracts to extract profits from good types, while still offering some separating contracts to extract profits from bad types.

Weak Adverse Selection: $\phi > 0$

Banks trade with both types (Pooling Equilibrium). Banks make more profits from pooling both firms at the reservation rate of good types when meeting captive firms. Consequently, the maximum rate observed in the market is $\bar{r^g}$, trades will only happen in the pooling region and all firms will receive credit. The CDF of rates is found by setting the equal profit condition on the pooling region:

$$[1-\pi+\pi(1-F^a(r))](r-c^{avg}(r)) = (1-\pi)(\bar{r^g}-c_{avg}(\bar{r^g}))$$

Proposition 5.3. *For $\pi \in (0,1)$, $\phi > 0$, $c'(r) < 1$ and with an uninformative signal, there exists a unique symmetric mixed-strategy equilibrium. Banks randomize their rate offers according to the cumulative distribution function $F(r)$ on the support $[\underline{r}, \bar{r}^g]$:*

$$F(r) = \begin{cases} 0 & \text{if } r < \underline{r} \\ 1 - \dfrac{1-\pi}{\pi}\left[\dfrac{(\bar{r}^g - r)}{(r - c^{avg}(r))}\right] & \text{if } \underline{r} \leq r \leq \bar{r}^g \\ 1 & \text{if } r > \bar{r}^g \end{cases}$$

Where the upper bound $\bar{r}^g$ is the reservation rate of the good type, and the lower bound $\underline{r}$ is implicitly defined by $F(\underline{r}) = 0$.

With mild adverse selection, the equilibrium is closest to the BJ dispersed price equilibrium, as banks offer only pooling contracts.

5.2.3 Equilibrium under informative signal

We now examine the scenario where banks can acquire an imperfect signal about each firm's type for a fixed cost C_s. This allows us to analyze the endogenous information acquisition decision and characterize the equilibrium as signal precision varies from uninformative to perfect.

Signal-Specific Equilibria. Conditional on acquiring the signal, banks face σ_g firms with good signals and σ_b firms with bad signals. For each group, the bank solves a rate-setting problem analogous to the no-information case, but with updated beliefs about the shares of types in the population receiving signal s_i. The sign of ϕ^{s_i} determines whether the equilibrium is pooling or separating. For each signal s_i, define:

$$\phi^{s_i} \equiv (\bar{r}^g - c^{avg,s_i}(\bar{r}^g)) - (1-\alpha^{s_i})\left(\bar{r}^b - c_b(\bar{r}^b)\right)$$

As before, the equilibrium is full pooling if $\phi^{s_i} > 0$, mixed if $-\frac{\pi}{1-\pi}\left(\bar{r}^g - c^{avg,s_i}(\bar{r}^g)\right) \leq \phi^{s_i} \leq 0$, fully separating if $\phi^{s_i} \leq -\frac{\pi}{1-\pi}\left(\bar{r}^g - c^{avg,s_i}(\bar{r}^g)\right)$

Bank profits. For firms with signal s_i, bank profits are:

$$\Pi^{s_i}(r) = \begin{cases} [1 - \pi + \pi(1 - F^{s_i}(r))] \, (r - c^{avg,s_i}(r)) & \text{if } r \in [\underline{r}^{s_i}, \bar{r}^g] \\ (1 - \alpha^{s_i}) \, [1 - \pi + \pi(1 - F^{s_i}(r))] \, (r - c_b(r)) & \text{if } r \in (\bar{r}^g, \bar{r}^b] \end{cases}$$

The first region corresponds to rates that both types (weighted by α^{s_i}) would accept; the second region contains only bad types. After observing signal s_i, the bank's expected pooling cost of lending at rate r is $c^{avg,s_i}(r) \equiv \alpha^{s_i} c_g(r) + (1 - \alpha^{s_i}) c_b(r)$, where $c_\theta(r) = r_{CB}/p_\theta(r)$ as before.

Proposition 5.4 (Signal Specific Equilibrium)**.** *For $\pi \in (0, 1)$ and under signal acquisition, there exists a unique symmetric mixed-strategy equilibrium. Conditional on receiving signal s_i, banks randomize their rate offers according to the cumulative distribution function $F^{s_i}(r)$ that takes the same functional form as in the no-information case, with population share μ_b replaced by $(1 - \alpha^{s_i})$ and average cost $c^{avg}(r)$ replaced by $c^{avg,s_i}(r)$. The aggregate distribution is the mixture of signal-specific distributions weighted by the shares of the population receiving the signal σ_i.*

Explicitly, the signal-specific distributions are as follows, depending on the sign of ϕ^{s_i}:

Case 1: Pooling equilibrium ($\phi^{s_i} > 0$)

$$F^{s_i}(r) = \begin{cases} 0 & \text{if } r < \underline{r}^{s_i} \\ 1 - \dfrac{1 - \pi}{\pi} \left[\dfrac{\bar{r}^g - r}{r - c^{avg,s_i}(r)} \right] & \text{if } \underline{r}^{s_i} \le r \le \bar{r}^g \\ 1 & \text{if } r > \bar{r}^g \end{cases}$$

where $\underline{r}^{s_i}$ solves $F^{s_i}(\underline{r}^{s_i}) = 0$.

Case 2: Hybrid/Separating equilibrium ($\phi^{s_i} \le 0$)

$$F^{s_i}(r) = \begin{cases} 0 & \text{if } r < \underline{r}^{s_i,a} \\ F^{s_i,a}(r) & \text{if } \underline{r}^{s_i,a} \le r \le \bar{r}^g \\ F^{s_i,a}(\bar{r}^g) + F^{s_i,b}(r) & \text{if } \bar{r}^g < r \le \bar{r}^b \\ 1 & \text{if } r > \bar{r}^b \end{cases}$$

where the equal-profit condition on each interval gives:

$$F^{s_i,a}(r) = \min\left\{1 - \frac{1-\pi}{\pi}\left[\frac{(1-\alpha^{s_i})(\bar{r}^b - c_b(\bar{r}^b))}{r - c^{avg,s_i}(r)} - 1\right], 0\right\}$$

$$F^{s_i,b}(r) = 1 - F^{s_i,a}(\bar{r}^g) - \frac{1-\pi}{\pi}\left[\frac{\bar{r}^b - c_b(\bar{r}^b)}{r - c_b(r)} - 1\right]$$

and the lower bounds $\underline{r}^{s_i,a}$, $\underline{r}^{s_i,b}$ are implicitly defined by $F^{s_i,a}(\underline{r}^{s_i,a}) = 0$ and $F^{s_i,b}(\underline{r}^{s_i,b}) = 0$.

Aggregate Rate Distribution. The distribution of rates across the entire population is the mixture of signal-specific distributions weighted by population fractions:

$$F(r) = \sigma_g F^{s_g}(r) + \sigma_b F^{s_b}(r)$$

The corresponding density is:

$$f(r) = \sigma_g f^{s_g}(r) + \sigma_b f^{s_b}(r)$$

where $f^{s_i}(r) = \frac{dF^{s_i}}{dr}$.

Convergence to Perfect Information

As signal precision approaches perfection ($q_g, q_b \to 1$), posterior beliefs converge to certainty:

$$\alpha^{s_g} = \frac{q_g\mu_g}{q_g\mu_g + (1-q_b)\mu_b} \to \frac{1\cdot\mu_g}{1\cdot\mu_g + 0\cdot\mu_b} = 1$$

$$\alpha^{s_b} = \frac{(1-q_g)\mu_g}{(1-q_g)\mu_g + q_b\mu_b} \to \frac{0\cdot\mu_g}{0\cdot\mu_g + 1\cdot\mu_b} = 0$$

where, as a reminder, α^{s_i} denotes the probability of being a good type given s_i. This implies:

- **Good signal group:** $c^{avg,s_g}(r) \to c_g(r)$, so

  $$\phi^{s_g} \to \bar{r}^g - c_g(\bar{r}^g) > 0$$

  The CDF $F^{s_g}(r)$ converges to $F_g(r)$, the good-type distribution under perfect information.

- **Bad signal group:** $c^{avg,s_b}(r) \to c_b(r)$, so

  $$\phi^{s_b} \to (\bar{r}^g - c_b(\bar{r}^g)) - \big(\bar{r}^b - c_b(\bar{r}^b)\big) \le 0$$

The CDF $F^{s_b}(r)$ converges to $F_b(r)$, the bad-type distribution under perfect information.

Furthermore, $\sigma_g \to \mu_g$ and $\sigma_b \to \mu_b$, so the aggregate distribution converges to:

$$F(r) \to \mu_g F_g(r) + \mu_b F_b(r)$$

which is the perfect-information equilibrium from Section 5.4.

Information Acquisition Decision

Proposition 5.5 (Signal Acquisition)**.** *Both banks acquire the signal if and only if the expected profit gain exceeds the cost:*

$$\mathbb{E}[\Pi^{signal}] - \mathbb{E}[\Pi^{no\ signal}] \geq C_s$$

where:

$$\mathbb{E}[\Pi^{signal}] = \sigma_g \cdot \Pi^{s_g} + \sigma_b \cdot \Pi^{s_b}$$

$$\mathbb{E}[\Pi^{no\ signal}] = \begin{cases} (1-\pi)(\bar{r}^g - c^{avg}(\bar{r}^g)) & \text{if } \phi > 0 \\ \mu_b(1-\pi)(\bar{r}^b - c_b(\bar{r}^b)) & \text{if } \phi \leq 0 \end{cases}$$

Comparative statics. The profit gain from acquiring information is larger when:

1. **Signal precision is high:** Higher q_θ increases separation between α^{s_g} and α^{s_b}
2. **Population heterogeneity:** Larger $|\mu_g - \mu_b|$ amplifies the value of distinguishing types
3. **Cost differences:** Larger $|c_g - c_b|$ increases the benefit of risk-based pricing
4. **Market power:** Lower π (less competition) increases profit margins, making information more valuable

In the limit of perfect competition ($\pi \to 1$), banks earn zero profits under both information regimes, so they will not pay any positive cost $C_s > 0$ to acquire information. This highlights the interaction between market structure and information acquisition incentives.

5.3 Welfare Implications

Total surplus. The population of firms aggregated for both banks is $2(1-\pi)+\pi$. Under perfect information or perfect signal, total surplus is realized as all firms are matched $TS^{q=1} = (2(1-\pi)+\pi)(\mu_b(\bar{r}^b - c_b) + \mu_g(\bar{r}^g - c_g))$. Under no information, total surplus depends on the severity of adverse selection: under partial or fully separating equilibrium, the share of good firms that is offered a rate beyond their reservation rate refuses and exits the market, thus the gains from trade remain unrealized.

Bank's and firms' share of total surplus. Under perfect information, the bank gets a share of realised surplus per trade exactly equal to $1-\pi$, while firms get the remaining π share. Under asymmetric information, banks profits are lower: in pooling, the bank looses $(1-\pi)(\bar{r^b} - \bar{r^g})$ from each bad type with respect to the perfect information case; in separating, the banks do not get the share of surplus from good firms that exit the market. In addition, under asymmetric information and pooling there is cross-subsidization between types: good types pay a share of the surplus equal to $(1-\pi)\mu_b(c_b - c_g)$. Asymmetric information constrains banks from achieving the profit levels under perfect information. When adverse selection is mild $\phi > 0$, the bank cannot fully extract surplus from bad firms due to pooling at lower interest rates. Conversely, when adverse selection is severe $\phi \leq 0$, the bank incurs losses as high-quality firms exit the market and low-quality firms are partially pooled at subsidized rates. Both cross-subsidization and exit are costs of asymmetric information that good firms are paying, and information reduces both costs. The following table illustrates the expected gains from trade for each bank:

		Perfect Info	**Imperfect Info**	**No Info**
Max surplus	$\phi > 0$	✓	✓	✓
	$\phi \leq 0$	✓	X	XX
Banks' profit loss	$\phi > 0$	–	↑	↑↑
	$\phi \leq 0$	–	↑	↑↑
Cross-subsidy	$\phi > 0$	–	↑	↑
	$\phi \leq 0$	–	↑ or –	↑ or –

Table 2: The table shows whether all gains from trade realize, the profit loss for banks and the degree of subsidization of good types to bad types under different information structures and values of ϕ.

6 Calibration

The next step is to give a quantitative assessment of the relative contribution of search and information to explained variance, along with the welfare trade-offs. Because the structural parameters, π for competition and q for the signal precision, are inherently latent, we employ a calibration exercise to recover these values by matching the model to specific moments of the residual distribution across Banque de France rating classes. For each rating class, we calibrate three parameters: the search friction intensity π, the signal precision q, and the reservation rate for good types $\bar{r}_g$. The central bank funding cost r_{CB} is fixed at 1, consistent with the average policy rate over the sample period. Four additional parameters are calibrated directly from the data: $\mu_b, p_g, p_b, \bar{r}_b$. We can observe the evolution of firms into good and bad rating classes; we can then denote as good/bad the firms that transition to better/worse rating classes, and derive the population share of bad types μ_b and the average success probabilities p_g and p_b in each population (computed as one minus the three-year-ahead default frequencies from Banque de France historical data), and the upper bound $\bar{r}_b$ (set at the maximum observed rate in each rating class). For each rating class, we rescale all residuals by the average rate over the period.

6.1 Calibration Strategy

The calibration proceeds in two stages. First, we conduct a grid search over the parameter space, evaluating 20^3 combinations of parameter values within economically plausible bounds: $\pi \in [0.05, 0.95]$, $q \in [0.5, 1.0]$, and $\bar{r}_g \in [0.5\,\bar{r}_{\text{mean}}, 0.98\bar{r}_b]$, where $\bar{r}_{\text{mean}}$ is the class mean rate. For each combination, we compute the model-implied moments—mean rate, variance, and correlation with subsequent default probability changes—using the equilibrium distribution derived in Section 5. We keep the combination that minimizes the objective function, which is the weighted sum of squared percentage errors of targeted moments. Second, we refine the best grid point using local optimization (L-BFGS-B algorithm) to minimize the weighted sum of squared percentage deviations between model and data moments.

The objective function to minimize is the sum of squared proportional errors across the three moments[8]:

$$Q(\theta) = \left(\frac{m_{\text{mean}}^{\text{model}} - m_{\text{mean}}^{\text{data}}}{m_{\text{mean}}^{\text{data}}}\right)^2 + \left(\frac{m_{\text{var}}^{\text{model}} - m_{\text{var}}^{\text{data}}}{m_{\text{var}}^{\text{data}}}\right)^2 + \left(\frac{m_{\text{corr}}^{\text{model}} - m_{\text{corr}}^{\text{data}}}{m_{\text{corr}}^{\text{data}}}\right)^2 \tag{5}$$

With three moments and three free parameters (r_{CB} fixed), the system is exactly identified. However, we impose additional structure through the equilibrium restrictions: the parameters must generate a feasible equilibrium (costs below reservation rates), and the distribution must satisfy the mixed-strategy conditions. These restrictions provide implicit identifying moment conditions beyond the three explicit moments we match.

6.2 Counterfactual Analysis

To decompose the relative contributions of search frictions and information acquisition, we compute counterfactual equilibria holding recovered parameters $(\hat{\pi}, \hat{\bar{r}}_g)$ and calibrated parameters $\mu_b, p_g, p_b, \bar{r}_b, r_{CB}$ fixed while varying the signal precision q. In particular, we compare the percentage increase in variance given the calibrated parameters against the counterfactual with $q = 0.5$ (no information) isolates the role of search frictions:

$$\text{Information Contribution} = \frac{\text{Var}(\hat{q}) - \text{Var}(q = 0.5)}{\text{Var}(q = 0.5)} \times 100\% \tag{6}$$

6.3 Calibration Results

Table 3 presents the calibrated parameters for each rating class; Table 4 reports the corresponding model fit. The correlation moment is matched closely across all classes. Mean and variance are less precisely matched, possibly reflecting unmodeled sources of dispersion.

[8]proportional errors are used to ensure that moments of different scales are weighted comparably in the estimation.

Table 3: Search Frictions and Information by Rating Class

Rating	$\hat{\pi}$	$\hat{q}$	$\hat{\bar{r}}_g$	Info Contr. (%)	Obj. Value $Q(\theta)$
3+	0.813	0.990	2.499	5.96	0.60
3	0.805	0.996	2.731	5.07	0.53
4+	0.788	0.993	3.434	4.48	0.36
4	0.789	1.000	3.605	4.53	0.42
5+	0.787	0.984	3.955	1.56	0.40
5	0.767	0.853	4.812	0.08	0.25
6	0.763	0.553	5.242	0.00	0.22
7	0.746	0.553	7.076	0.00	0.12

Notes: $\hat{\pi}$ is the search-friction parameter, $\hat{q}$ the signal precision, $\hat{\bar{r}}_g$ the reservation rate of good borrowers, and "Information Contribution" measures the percentage increase in rate dispersion attributable to screening relative to the no-information benchmark.

Table 4: Model Fit: Observed vs. Model-Implied Moments

	Mean		Variance		Correlation	
Rating	Data	Model	Data	Model	Data	Model
3+	1.67	1.59	0.400	0.090	0.025	0.025
3	1.83	1.70	0.440	0.119	0.033	0.033
4+	2.40	2.04	0.550	0.231	0.032	0.032
4	2.37	2.12	0.730	0.264	0.048	0.047
5+	2.57	2.27	0.860	0.326	0.040	0.040
5	3.36	2.71	0.980	0.527	0.017	0.017
6	3.63	2.92	1.130	0.652	0.000	0.000
7	4.79	3.88	1.840	1.306	0.000	0.000

Notes: All moments computed on residual rates within each rating class. Correlation is between the residual rate and the three-year-ahead default indicator.

The calibration results reveal three key findings. First, search frictions are relatively low across all rating classes, with $\hat{\pi}$ estimated between $[0.74, 0.81]$, increasing over ex-ante safety. This suggests that the French corporate lending market is relatively competitive, with most firms sampling multiple lenders—consistent with the high concentration of bank relationships documented in Section 4.1, where on average 60% of firms across classes maintain at least two active banking relationships.

Second, screening intensity exhibits heterogeneity across the credit quality distribution. Good borrowers (ratings 3+ through 5+) face near-perfect screening ($\hat{q} \geq 0.98$), while poor borrowers (ratings 6 and 7) face essentially no screening ($\hat{q} \approx 0.55$).

Third, despite high screening intensity implied by the data moments for good ratings, information contributes minimally to observed rate dispersion. The information contribution to variance never exceeds 5%, indicating that search frictions account for most of unexplained dispersion even in segments with near-perfect screening. This partly reflects the mild severity of the adverse selection problem within each rating class, for example, the relative low distance of default probabilities between types mechanically limit between-type variance.

6.4 TBA: example of welfare loss/surplus share of banks/firms, cross-subsidy between good and bad types, etc.

in progress

7 Monetary policy implications

This section examines the implications of the model for monetary policy transmission via the bank lending channel. Specifically, we analyze how market power and information asymmetries affect the pass-through of policy rate changes to lending rates.

7.1 Model predictions

The model yields key implications on monetary policy transmission.

1. Low competition impairs monetary policy transmission As expected, less competition impairs the pass-through of a rate increase. When competition is low, banks possess market power; they set rates well beyond marginal cost and close to borrowers' reservation rates. In this environment, banks would only partially pass-through a policy rate hike, to preserve both their market share and their profit margins. In sum, following a rate hike, we expect the average rate to be stickier in a setting where the market is both non-competitive and we should expect lower rate dispersion around the average.

2. The information structure interacts with competition. Bank's screening ability and its role in loan pricing is a critical determinant of policy rates transmission. When banks fail to screen, the pass-through is strengthened/weakened, depending on actual payoffs.

3. Monetary policy affects gains from trade in the market. Higher rates reduce gains from trade in the market.

This in turn increases the severity of adverse selection. Thus an increase in policy rates in a market with little screening may determine the transition from a pooling to a screening strategy by banks, with the exit of firms with lower expected returns.

7.2 Empirical evidence: Rate dispersion responds to monetary policy

We test the model implications by examining the response of the distribution of interest rate residuals (i.e. the unexplained term) from the baseline regression to monetary policy. We compute high-frequency monetary policy shocks for the euro area, following the methodology of Altavilla et al. (2019). These shocks are identified using changes in asset prices (OIS) around ECB monetary policy announcements, capturing the unexpected component of policy decisions. The identifying assumption is that monetary policy, which is endogenous to asset price changes, would not respond at such a high frequency to changes in asset prices; in fact, the short window around the announcement ensures to capture the market surprise. The used shock series is reported in Fig. B.8 in Appendix.

First, we expect a monetary policy shock to reduce the variance and skewness of the observed rate distribution. We regress the variance and skewness of loan residuals on the shocks. The results indicate that both dispersion (variance) and asymmetry (skewness) of loan rate residuals respond systematically to monetary policy surprises, suggesting that policy changes influence not only the average cost of borrowing, but also the degree of heterogeneity across firms. The specification is carried out in both levels and changes. In levels:[9]

$$Y_{lender,t} = \beta_0 + \beta_1 \cdot \text{MPshock}_t + \beta_2 Y_{lender,t-1} + \text{FE}_{lender} + \varepsilon$$

And in first differences:[10]

$$\Delta Y_{lender,t} = \gamma_0 + \gamma_1 \cdot \text{MPshock}_t + \text{FE}_{lender} + \epsilon$$

[9] β_1 measures the effect of MP on the level of dispersion of premia , controlling for its lag. Assumes that there is inertia in the dispersion of rates over time.

[10] γ_1 measures the effect of MP on the change of dispersion of premia. No need to worry about trend.

	Dispersion		ΔDispersion		Skewness		ΔSkewness	
L1.rate shock	-0.0096***	-0.0085***	-0.0071**	-0.0049	-0.0135***	-0.0099*	0.0012	0.0053
	(0.000)	(0.003)	(0.028)	(0.175)	(0.010)	(0.093)	(0.860)	(0.490)
L2.rate shock		0.0001		0.0069**		-0.0227***		-0.0024
		(0.979)		(0.041)		(0.000)		(0.732)
L3.rate shock		-0.0017		-0.0007		-0.0208***		-0.0099
		(0.568)		(0.851)		(0.000)		(0.196)
L1.SD / L1.Skew	0.2813***	0.2799***			0.1294***	0.1171***		
	(0.000)	(0.000)			(0.000)	(0.000)		
Lender FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	7,189	7,077	7,189	7,077	7,189	7,077	7,189	7,077
R^2	0.40	0.41	0.01	0.01	0.23	0.24	0.00	0.00

p-values in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table 5: Data: Nouveaux Credits, BdF, 2006q1-2024q2, quarterly shocks. The table shows the effect of a monetary policy rate shock on the standard deviation and skewness of the residuals' distribution.

Table 5 show that a monetary tightening has a downward effect on the variance of the distribution of the unexplained component of interest rates, both in levels and changes, and a negative effect on the level of the skewness. This is consistent with what is expected in the theoretical framework. Further analysis on the response in different markets with varying competition/screening levels would further clarify the strength of the pass-through with respect to the competitive and perfect screening benchmark.

8 Conclusion

In this paper, we explore the extent to which information acquisition and imperfect competition jointly determine loan prices. We empirically document interest rate dispersion and use it to derive important relationships in the data, showing that rates contain forward information about the future evolution of firms' creditworthiness. Specifically, there is a positive and significant correlation between paying a relatively higher rate and being ex-post more likely to default.

To shed light on these dynamics, we propose a framework where loan pricing is the outcome of imperfect competition and asymmetric information. Our theoretical analysis yields two main results: first, imperfect competition delivers within-type dispersion; second, the information structure determines between-type dispersion when banks acquire additional information.

The model yields key implications on monetary policy transmission. As expected, limited competition generally weakens the pass-through of policy rates. Furthermore, the interaction of search frictions and weak information acquisition creates a specific rigidity in rates that further impairs the transmission of policy shocks. Using high-frequency monetary policy shocks, we find a reduction in the dispersion of interest rate residuals, consistent with the model predicts that total observed dispersion decreases following an increase in the policy rate.

References

Agarwal, S., Grigsby, J., Hortaçsu, A., Matvos, G., Seru, A., and Yao, V. (2024). Searching for approval. *Econometrica*, 92(4):1195–1231.

Agarwal, S. and Hauswald, R. (2010). Distance and private information in lending. *The Review of Financial Studies*, 23(7):2757–2788.

Akerlof, G. A. (1970). The market for lemons: Quality uncertainty and the market mechanism. *The Quarterly Journal of Economics*, 84(3):488–500.

Altavilla, C., Brugnolini, L., Gürkaynak, R. S., Motto, R., and Ragusa, G. (2019). Measuring euro area monetary policy. *Journal of Monetary Economics*, 108:162–179.

Altavilla, C., Gürkaynak, R., and Quaedvlieg, R. (2024). Macro and micro of external finance premium and monetary policy transmission. *Journal of Monetary Economics*, 147, Supplement.

Amiti, M., Kashyap, A. K., and Kovner, A. (2026). Why do firms pay different interest rates on their bank loans? *Working paper*.

Bech, M. and Monnet, C. (2015). A search-based model of the interbank money market and monetary policy implementation. *BIS Working Papers No 529*.

Berger, A. N. and Udell, G. F. (1995). Relationship lending and lines of credit in small firm finance. *The Journal of Business*, 68(3):351–381.

Berger, A. N. and Udell, G. F. (2002). Small business credit availability and relationship lending: The importance of bank organisational structure. *The Economic Journal*, 112(477):F32–F53.

Bernanke, B. S. and Gertler, M. (1995). Inside the black box: the credit channel of monetary policy transmission. *Journal of Economic Perspectives*, 9(4):27–48.

Boot, A. W. A. (2000). Relationship banking: What do we know? *Journal of Financial Intermediation*, 9(1):7–25.

Boot, A. W. A. and Thakor, A. V. (2000). Can relationship banking survive competition? *The Journal of Finance*, 55(2):679–713.

Burdett, K. and Judd, K. L. (1983). Equilibrium price dispersion. *Econometrica*, 51(4):955–969.

Cahn, C., Girotti, M., and Salvadè, F. (2024). Credit ratings and the hold-up problem in the loan market. *Management Science*.

Carletti, E., Leonello, A., and Marquez, R. (2024). Market power in banking. *ECB Working Paper*, (2886).

Cavalcanti, T., Kaboski, J., Martins, B., and Santos, C. (2021). Dispersion in financing costs and development. *NBER Working Paper Series*, (28635).

Claessens, S., Ongena, S., and Wang, T. (2025). ”if you don’t know me by now ...” information collection by banks in lending to private firms. *Working Paper*.

Core, F., De Marco, F., Eisert, T., and Schepens, G. (2025). Inflation and floating-rate loans: Evidence from the euro-area. *CEPR Discussion Paper*, (20354).

Crawford, G. S., Pavanini, N., and Schivardi, F. (2018). Asymmetric information and imperfect competition in lending markets. *American Economic Review*, 108(7):1659–1701.

Degryse, H. and Ongena, S. (2005). Distance, lending relationships, and competition. *The Journal of Finance*, 60(1):231–266.

Delis, M., Ferrando, A., Mulier, K., and Ongena, S. (2025). The poor, the rich, and the credit channel of monetary policy. ECB Working Paper 3058, European Central Bank.

Dell’Ariccia, G. and Marquez, R. (2006). Lending booms and lending standards. *The Journal of Finance*, 61(5):2511–2546.

Den Haan, W. J., Ramey, G., and Watson, J. (2003). Liquidity flows and fragility of business enterprises. *Journal of Monetary Economics*, 50(6):1215–1241.

Drechsler, I., Savov, A., and Schnabl, P. (2017). The deposits channel of monetary policy. *The Quarterly Journal of Economics*, 132(4):1819–1876.

Duffie, D., Gârleanu, N., and Pedersen, L. H. (2005). Over-the-counter markets. *Econometrica*, 73(6):1815–1847.

Farboodi, M., Jarosch, G., Menzio, G., and Wiriadinata, U. (2025). Intermediation as rent extraction. *Journal of Economic Theory*, 227:106029.

Flanagan, T. (2023). The value of bank lending. *Journal of Finance*. Forthcoming.

Guerrieri, V., Shimer, R., and Wright, R. (2010). Adverse selection in competitive search equilibrium. *Econometrica*, 78(6):1823–1862.

Gürkaynak, R., Karasoy-Can, H. G., and Lee, S. S. (2022). Stock market’s assessment of monetary policy transmission: The cash flow effect. *The Journal of Finance*, 77(4):2375–2421.

Hauswald, R. and Marquez, R. (2006). Competition and strategic information acquisition in credit markets. *The Review of Financial Studies*, 19(3):967–1000.

Holmström, B. and Tirole, J. (1997). Financial intermediation, loanable funds, and the real sector. *The Quarterly Journal of Economics*, 112(3):663–691.

Ioannidou, V. and Ongena, S. (2010). "time for a change": Loan conditions and bank behavior when firms switch banks. *The Journal of Finance*, 65(5):1847–1877.

Jiménez, G., Ongena, S., Peydró, J.-L., and Saurina, J. (2012). Credit supply and monetary policy: Identifying the bank balance-sheet channel with loan applications. *The American Economic Review*, 102(5):2301–2326.

Jiménez, G., Ongena, S., Peydró, J.-L., and Saurina, J. (2014). Hazardous times for monetary policy: What do twenty-three million bank loans say about the effects of monetary policy on credit risk-taking? *Econometrica*, 82(2):463–505.

Kashyap, A. K. and Stein, J. C. (2000). What do a million observations on banks say about the transmission of monetary policy? *The American Economic Review*, 90(3):407–428.

Kirti, D. (2020). Why do bank-dependent firms bear interest-rate risk? *Journal of Financial Intermediation*, 41:100811.

Kozeniauskas, N. (2024). Beyond risk: Firm financing and interest rates. *Working Paper*.

Lester, B., Shourideh, A., Venkateswaran, V., and Zetlin-Jones, A. (2019). Screening and adverse selection in frictional markets. *Journal of Political Economy*, 127(1):338–377.

Mazet-Sonilhac, C. (2025). Information frictions in credit markets. *Working Paper*.

Petersen, M. A. and Rajan, R. G. (1995). The effect of credit market competition on lending relationships. *The Quarterly Journal of Economics*, 110(2):407–443.

Puglisi, T. (2023). Retail credit markets and the passthrough of monetary policy. *Working Paper*.

Rajan, R. G. (1992). Insiders and outsiders: The choice between informed and arm's-length debt. *The Journal of Finance*, 47(4):1367–1400.

Scharfstein, D. and Sunderam, A. (2016). Market power in real estate. *Working Paper*.

Sharpe, S. A. (1990). Asymmetric information, bank lending, and implicit contracts: A stylized model of customer relationships. *The Journal of Finance*, 45(4):1069–1087.

Stiglitz, J. E. and Weiss, A. (1981). Credit rationing in markets with imperfect information. *The American Economic Review*, 71(3):393–410.

Varian, H. R. (1980). A model of sales. *The American Economic Review*, 70(4):651–659.

Wang, Z., He, Z., and Huang, J. (2022). Monetary policy, bank competition, and corporate investment. *Journal of Finance*, 77(5):2575–2615.

Wasmer, E. and Weil, P. (2004). The macroeconomics of labor and credit market imperfections. *The American Economic Review*, 94(4):944–963.

A Additional tables and figures

LA COTE DE CRÉDIT
(capacité de l'entreprise à honorer ses engagements financiers à 3 ans)

3++	Excellente
3+	Très forte
3	Forte
4+	Assez forte
4	Correcte
5+	Assez faible
5	Faible
6	Très faible
7	Appelant une attention spécifique présence d'au moins un incident de paiement significatif
8	Menacée
9	Compromise
P	Procédure collective redressement ou liquidation judiciaire
0	Pas de documentation comptable analysée et absence d'informations défavorables

Figure A.1: Source: Banque de France. Credit rating scale until 2022.

B Additional facts on rate dispersion

Table B.1: Data: Nouveaux Credits (BdF), 2006q1-2024q2. Excludes credit lines and personal loans. All credit and balance-sheet variables are expressed in terms of reported assets.

	Total effective rate					
log.Loan quantity (K-Eur)	-0.2219***	-0.1535***	-0.1537***	-0.1436***	-0.1467***	-0.0989**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Maturity (months)	-0.0005*	0.0022***	0.0032***	0.0021*	0.0022***	0.0020***
	(0.073)	(0.002)	(0.001)	(0.003)	(0.003)	(0.004)
Age	-0.0141***	-0.0095***	-0.0084***	-0.0074***	-0.0073***	-0.0038***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Age squared	0.0001***	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Young firm 0-3y (dummy)	-0.0577***	-0.0404	-0.0424	-0.0190	-0.0161	-0.0763**
	(0.001)	(0.297)	(0.285)	(0.528)	(0.575)	(0.012)
New bank-firm relationship (dummy)	0.1988***	0.0183	0.0145	-0.0171	-0.0179	-0.0339**
	(0.000)	(0.639)	(0.715)	(0.293)	(0.285)	(0.016)
Variable rate (dummy)	0.4528***	0.4905***	0.1665*	0.1184	0.1167	0.1434
	(0.000)	(0.000)	(0.051)	(0.196)	(0.196)	(0.119)
Personal loan (dummy)	-0.2999**	-0.0612	0.0400	-0.0066	-0.0043	0.0603
	(0.035)	(0.674)	(0.620)	(0.943)	(0.962)	(0.447)
Individual entrepreneur (dummy)	0.5724***	0.5680***	0.5333***	0.4320***	0.4195***	0.2404***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Loan over assets	0.1613**	0.0983*	0.0986*	0.1057*	0.0988	0.0404
	(0.021)	(0.064)	(0.080)	(0.079)	(0.076)	(0.061)
Pledged guarantees	0.8325***	0.1146	-0.0204	-0.2563**	-0.2360**	-0.1759*
	(0.003)	(0.132)	(0.709)	(0.016)	(0.011)	(0.025)
Revenues	-0.0444***	-0.0165	-0.0114	0.0034	0.00059	-0.009
	(0.000)	(0.367)	(0.367)	(0.592)	(0.007)	(0.006)
Bank Debt	0.1594***	0.1219**	0.1158**	0.1097***	0.1616***	-0.0168
	(0.000)	(0.008)	(0.014)	(0.001)	(0.001)	(0.017)
Non-Bank Debt	-0.0395	-0.0583	-0.0455	-0.0055	-0.0009	-0.0167
	(0.401)	(0.630)	(0.723)	(0.945)	(0.991)	(0.742)
Mobilised credit	-0.0117*	-0.0021	0.0013	0.0000	0.0003	0.0025
	(0.023)	(0.471)	(0.700)	(0.991)	(0.927)	(0.300)
Short Term Debt	0.0009	-0.0024	-0.0009	0.0190	0.0190	-0.0125
	(0.967)	(0.871)	(0.951)	(0.159)	(0.159)	(0.283)
Undrawn credit lines (K-Eur)	-0.0109	-0.0058	-0.0044	-0.0070	-0.0071	-0.0025
	(0.967)	(0.871)	(0.951)	(0.159)	(0.159)	(0.283)
multi-loan (K-Eur)	-0.2185***	-0.0686	-0.0949	-0.0513	-0.0403	-0.0210
	(0.068)	(0.0605)	(0.058)	(0.036)	(0.034)	(0.022)
multi-product (K-Eur)	-0.0327	-0.0140	-0.0470	-0.0389	-0.0376	-0.0181
	(0.022)	(0.029)	(0.027)	(0.021)	(0.018)	(0.011)
Quarter FE	✓					
Lender × Quarter FE		✓				
Lender × Quarter × Loan Type FE			✓			
Lender × Quarter × Loan Type × 2digit Sector FE				✓	✓	✓
Lender × Quarter × Dep.code FE					✓	✓
Group, Rating, Size FE						✓
Observations	417,953	417,953	417,953	417,953	417,953	417,953
R^2	0.55	0.72	0.76	0.82	0.85	0.86

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

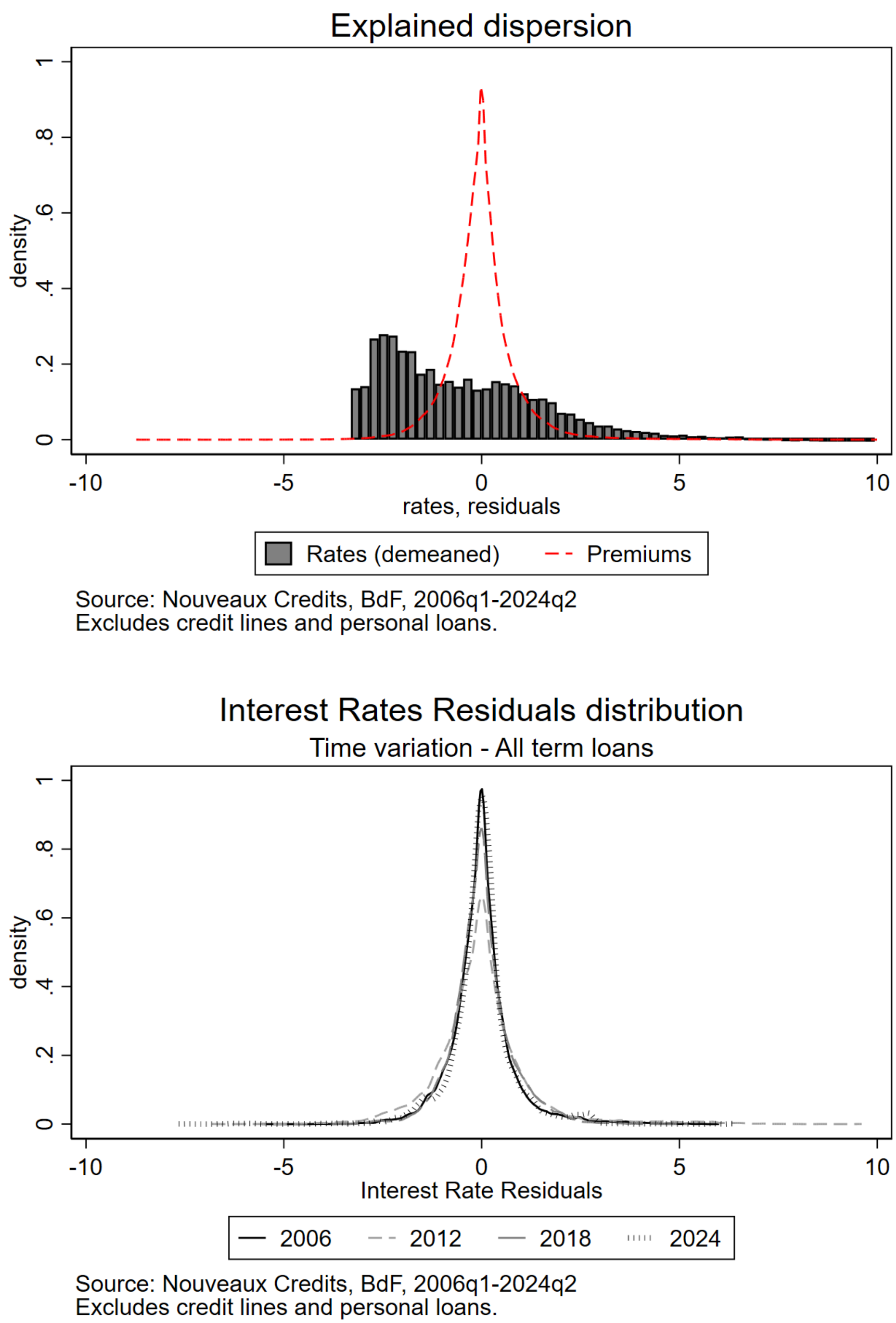
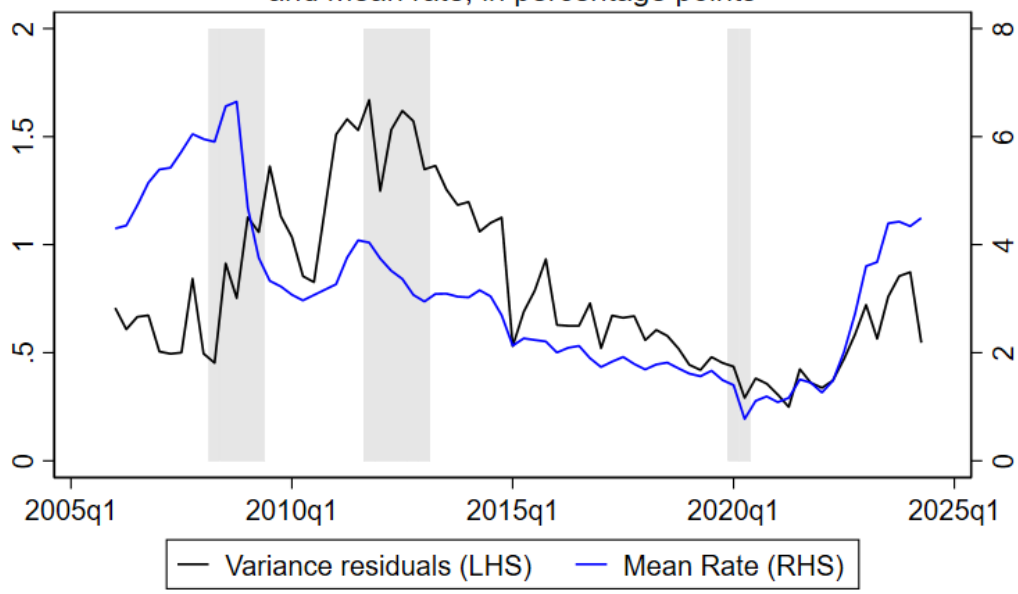


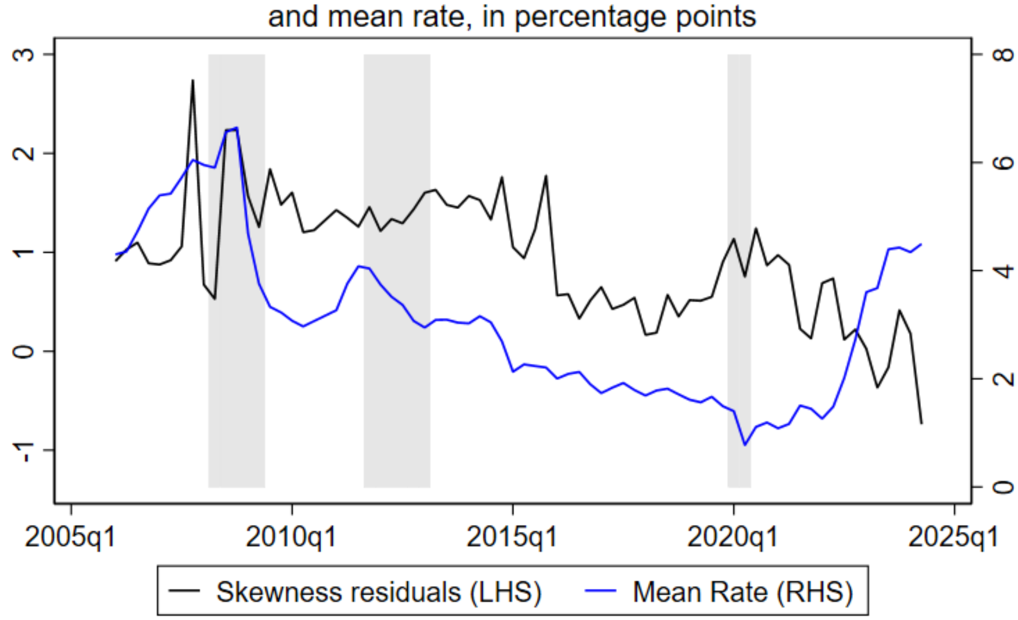
Figure B.2: Regression outcome. Top: residuals versus actual rates, all sample. Bottom: residuals distribution for selected years.

Variance of Interest rates premia
and mean rate, in percentage points



Source: Nouveaux Credits, BdF, 2006q1-2024q2. All term loans.
Recession dates for France from AFSE: 2008q2-2009q2, 2011q4-2013q1, 2020q1-2020q2

Skewness of Interest rates premia
and mean rate, in percentage points



Source: Nouveaux Credits, BdF, 2006q1-2024q2. All term loans.
Recession dates for France from AFSE: 2008q2-2009q2, 2011q4-2013q1, 2020q1-2020q2

B.1 Additional specification 1: Firm fixed effects

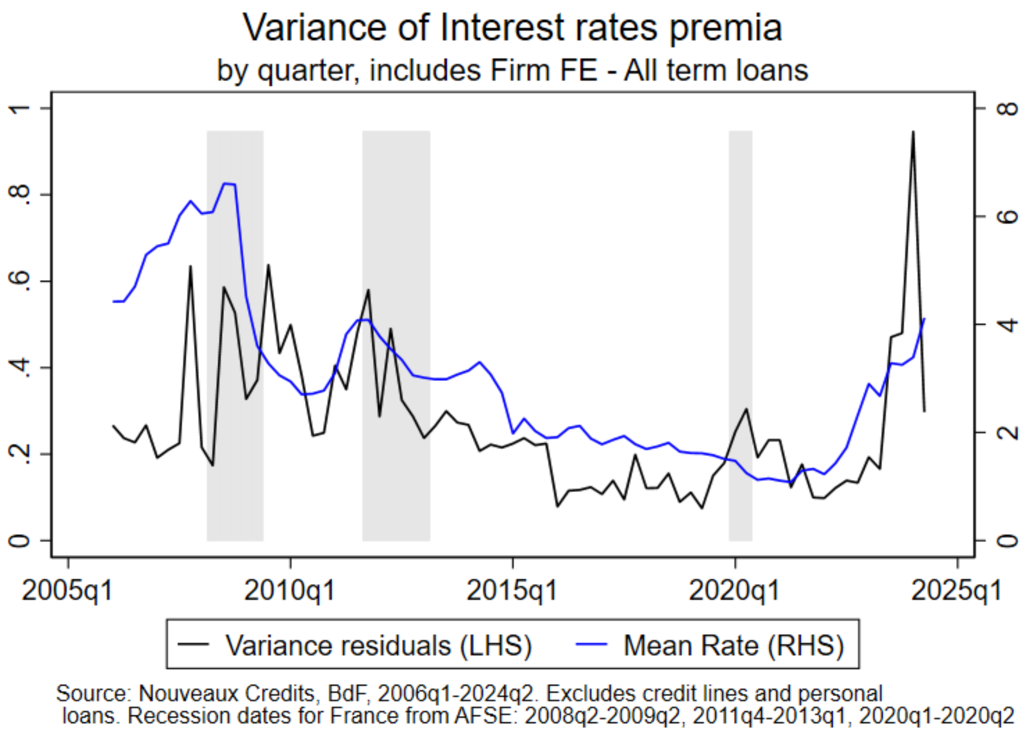
	Total effective rate					
Loan quantity (K-Eur)	-0.0000***	-0.0000***	-0.0000	-0.0000	-0.0000**	-0.0000
	(0.000)	(0.006)	(0.486)	(0.143)	(0.044)	(0.117)
Maturity (months)	-0.0016***	0.0013***	0.0015***	0.0015***	0.0014***	0.0014***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Variable rate (dummy)	0.2381***	0.1357***	0.2176***	0.2368***	0.2113***	0.1995***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Loan over assets	0.0524***	-0.0068**	-0.0076***	-0.0070**	-0.0069**	-0.0074***
	(0.000)	(0.025)	(0.009)	(0.012)	(0.012)	(0.007)
Quarter FE	✓					
Firm × Quarter FE		✓	✓	✓	✓	
Lender FE			✓			
Lender × Quarter FE				✓	✓	
Firm × Lender × Quarter FE						✓
Loan Type FE					✓	✓
Observations	590,621	197,343	197,328	196,048	196,042	156,211
R^2	0.51	0.92	0.93	0.94	0.94	0.94

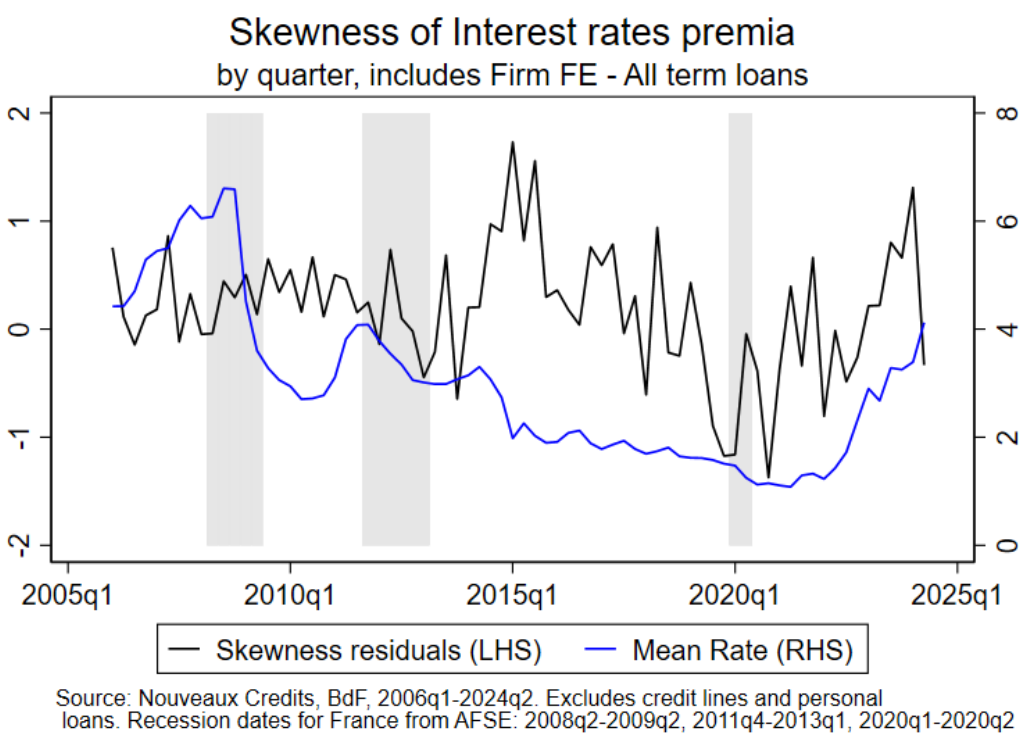
p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table B.2: "Data: Nouveaux Credits (BdF), 2006q1-2024q2""Excludes credit lines and personal loans. All credit and"" balancesheet variables are expressed in terms of reported assets."

Figure B.3: variance and skewness of residuals from firm FE specification





B.2 Additional specification 2: truncated regression approach

One possible concern is that it is not possible to observe rates below 0. Regression approaches assuming normality may be downward biased, predicting negative rates even when they are impossible to observe. Two approaches can help address this issue. Firstly, one can verify to what extent the OLS is predicting negative values. It seems that most of the variation being absorbed by fixed effect, this is a minor concern, as 0.4% of predictions fall slightly below zero. Secondly, the same analysis can be repeated using Tobit regression, which accounts for this lower bound; this approach is more limited towards the inclusion of many interactions of fixed effects, thus it forces a more parsimonious approach. In any case, the resulting distribution of residuals does not seem very affected.

B.3 Rate dispersion and bank heterogeneity

One question that naturally arises is: is this dispersion hiding heterogeneity across banks? The answer is yes. There is consistent heterogeneity between banks in the explained variation, as shown below by repeating the above estimation of rates by bank, and collecting the R^2.

$$\text{Rate}_{i,b,t} = \alpha + \beta \cdot \text{LoanChar}_{i,b,t} + \gamma \cdot \text{FirmChar}_{i,t} + \lambda_t + \varepsilon_{i,b,t} \tag{7}$$

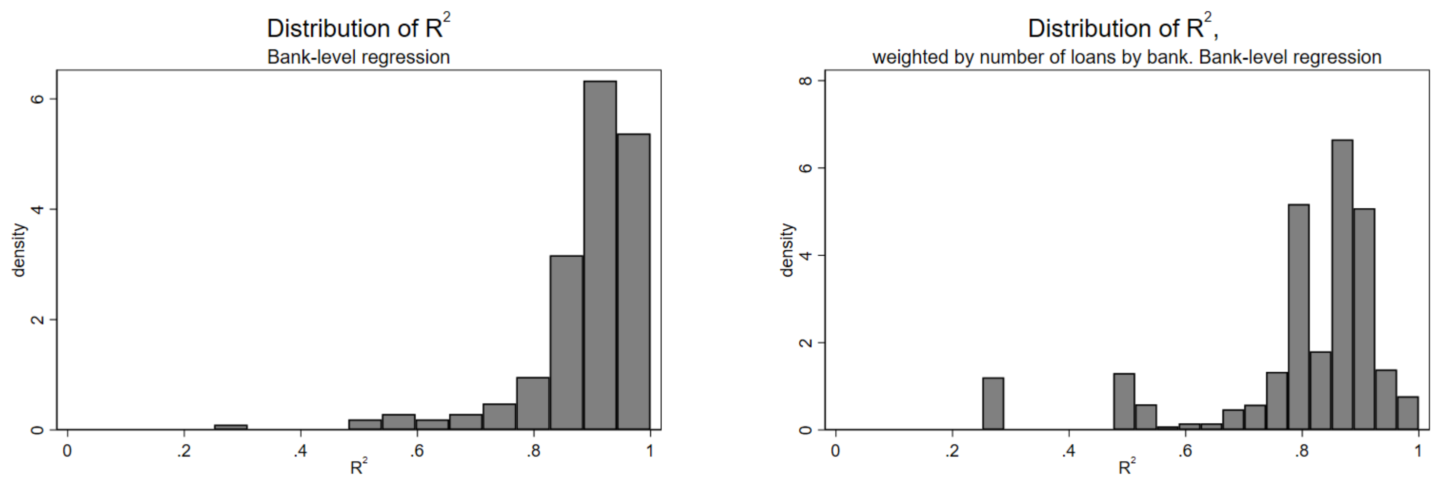


Figure B.4: The right image reports the distribution of R^2 weighted by the number of loans issued.

In the following section the credit rating of firms will be used as a source of public information available to banks and to study the evolution of firm's outcome in terms of their default probability/outcome. Consequently, it is interesting at this stage to report to which extent banks use these ratings. Firstly, the same estimation is repeated by including only the rating as explanatory variable, along with time and loan type FE and by lender, and collect the R^2.

$$\text{Rate}_{i,b,t} = \alpha + \beta \cdot \text{Firm Rating}_{i,b,t} + \lambda_t + \varepsilon_{i,b,t} \tag{8}$$

where the only fixed effects kept are by loan type and quarter.

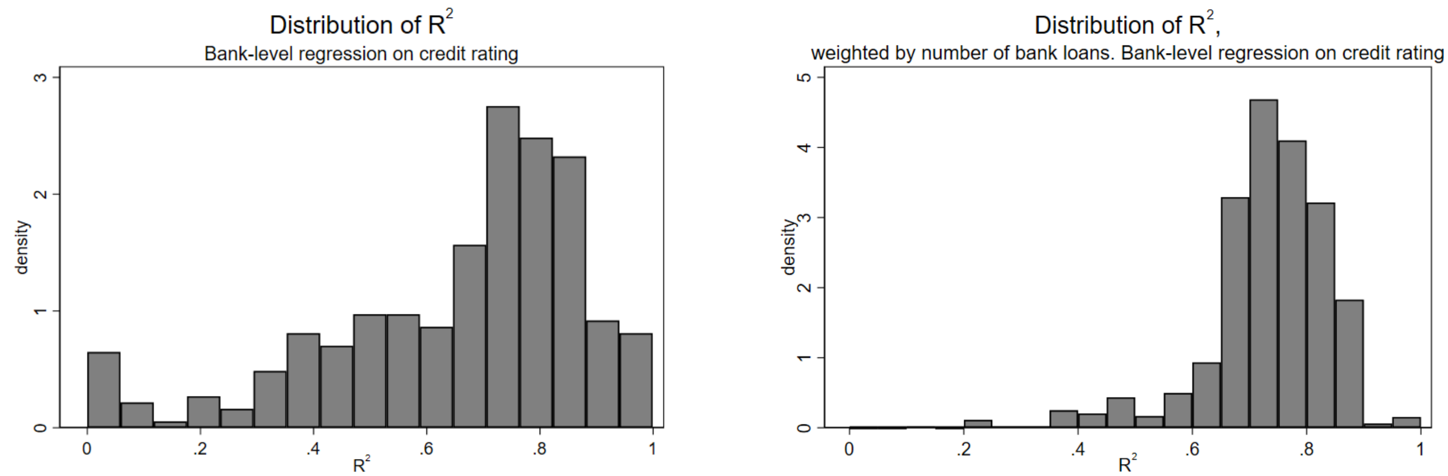


Figure B.5: The right image reports the distribution of R^2 weighted by the number of loans issued.

Secondly, one can look at how much of the variation is explained when including the public rating, in particular by collecting the increment in the R^2.

$$\begin{aligned}
\text{Rate}_{i,b,t} &= \alpha + \lambda_t + \varepsilon_{1,i,b,t} && \rightarrow \text{Get } R_1^2 \\
\text{Rate}_{i,b,t} &= \alpha + \beta \cdot \text{Firm Rating}_{i,b,t} + \lambda_t + \varepsilon_{2,i,b,t} && \rightarrow \text{Get } R_2^2 \\
\Delta R^2 &= R_2^2 - R_1^2
\end{aligned}$$

where the only fixed effects kept are by loan type and quarter.

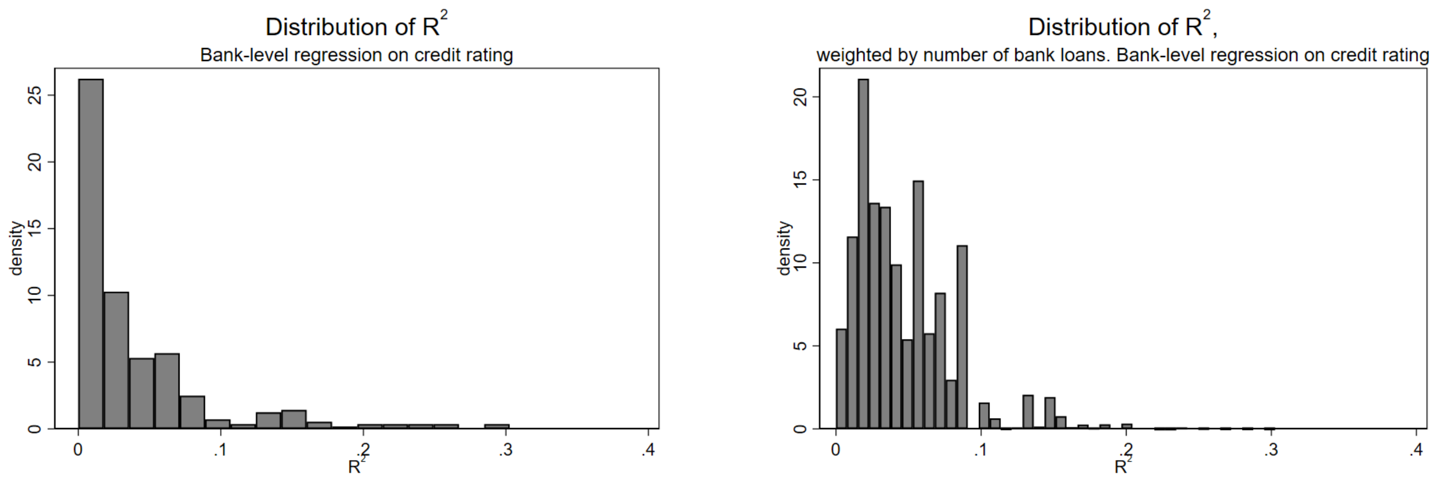


Figure B.6: The right image reports the distribution of R^2 weighted by the number of loans issued.

B.4 Residuals and market competition

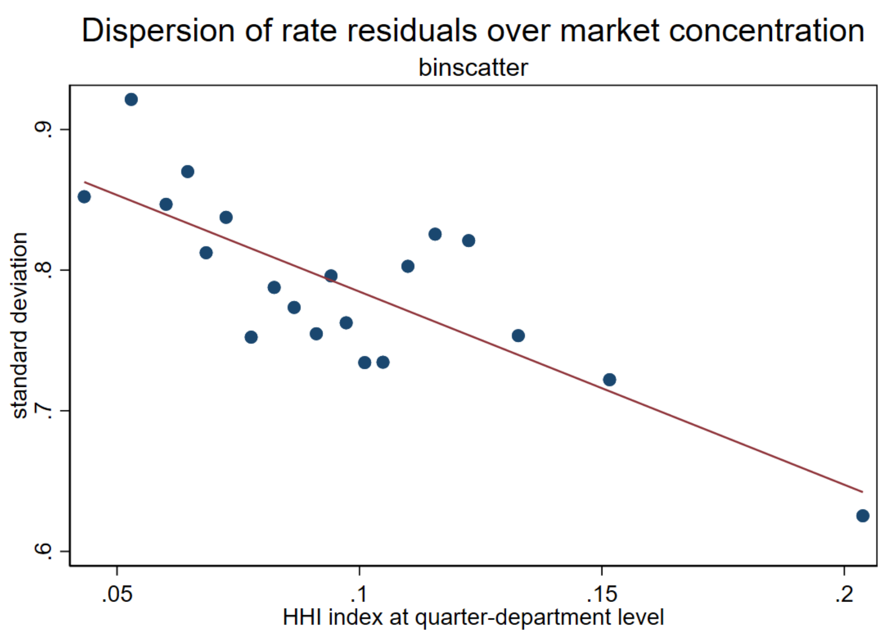
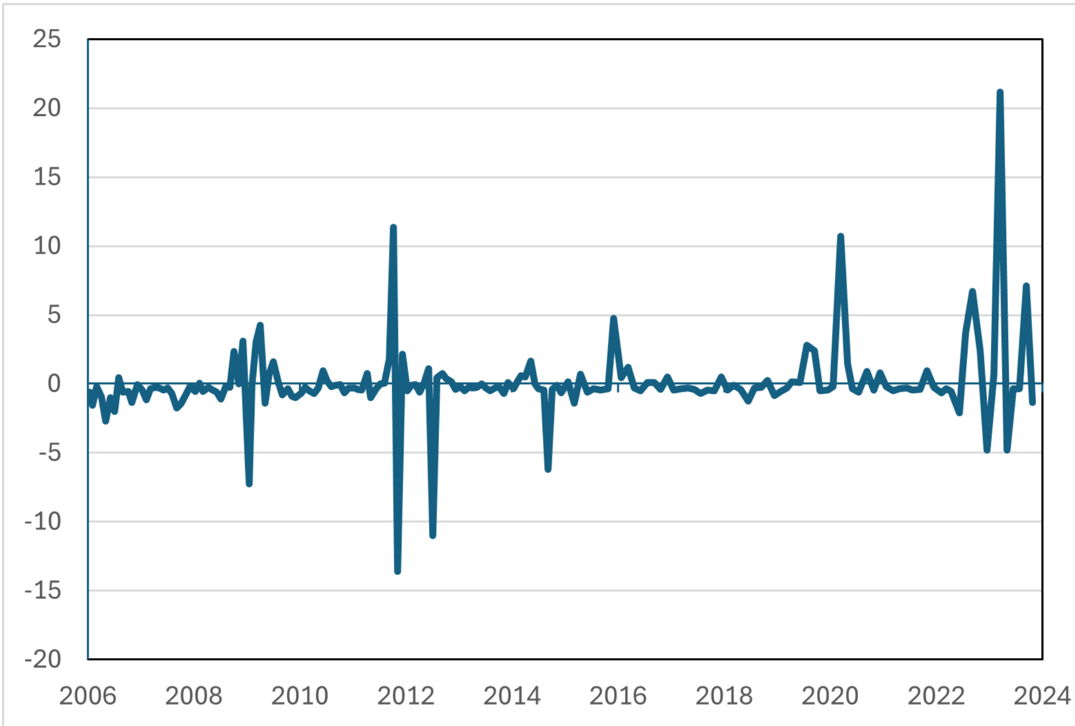


Figure B.7: Unexplained rate residuals' dispersion over HHI by department-quarter.

B.5 Monetary policy implications

Figure B.8: Interest rate shock factor estimated from changes in short term OIS rates in a narrow window after the ECB press release.



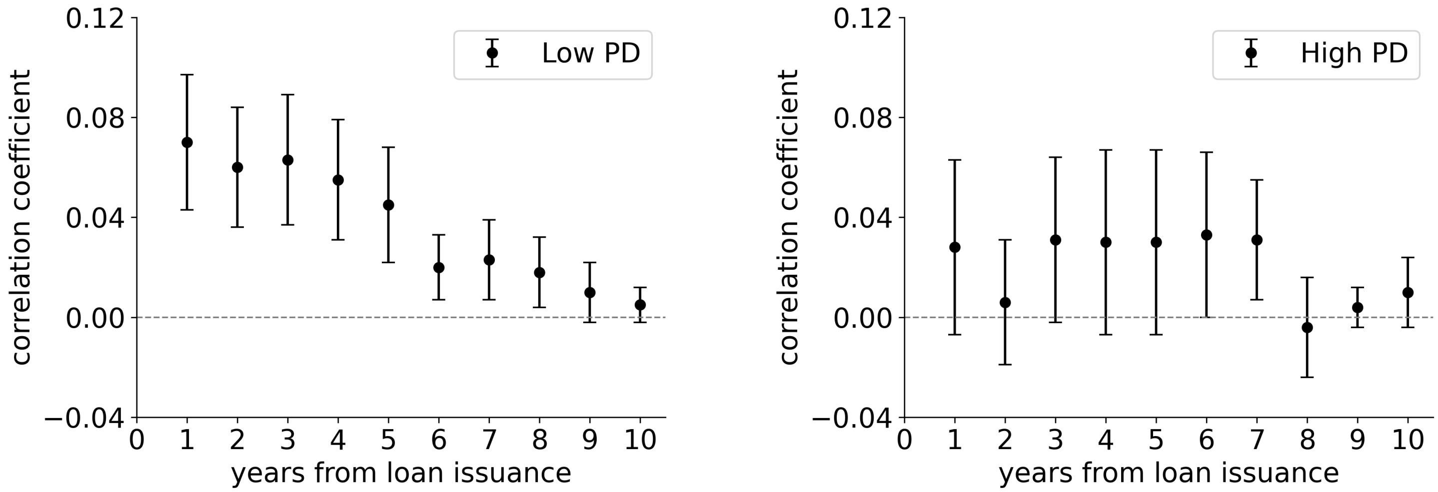
B.6 Excluding the mechanical effect of higher rates on default probability

We adopt two approaches to show that the observed correlation between rate residuals and unexpected changes in riskiness is not driven by a leverage effect - higher rates induce higher ex-post riskiness through the worsening of firm's fundamentals. Instead, it identifies banks' additional information with respect to the econometrician - banks recognize riskier types within the same rating class and offer a higher rate. A more comprehensive treatment of this analysis is provided in the appendix.

Approach 1: analyzing the correlation in the cross section.

We study how the correlation varies across different rating classes and firm characteristics to test if the correlation we observa can be potentially driven by a mechanical effect of rates on firm fundamentals. One first hypothesis is that the leverage effect should be stronger for firms that are already closer to the default threshold, namely in worse rating classes. We find the opposite, namely a stronger effect in ex-ante safer classes, while it becomes non significant for worse ratings (Fig. B.9).

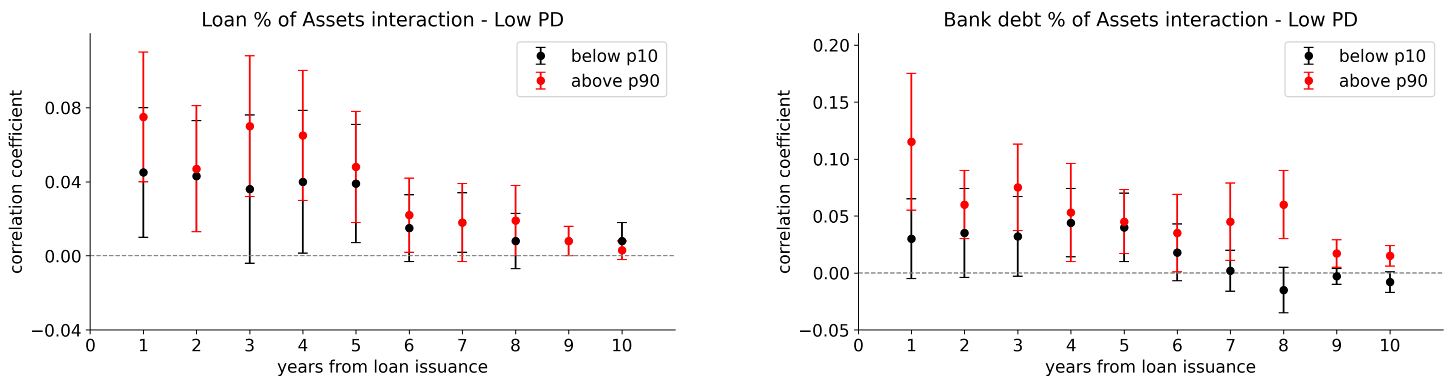
Figure B.9: Correlation computed separately for firms with ex ante good rating (default probability $< 0.62\%$) vs bad (default probability $> 3.45\%$).



The second hypothesis is that, if the deterioration in firm fundamentals was solely the result of receiving a worse rate offer ex-ante, the effect should be stronger when the loan represents a larger share of the firm's balance sheet. We test whether loan size relative to the firm's balance sheet explains the correlation between higher ex-ante rates and higher ex-post default. Empirically, we compare firms with a relatively large share of loan over asset (above 90^{th} percentile) and firms with a relatively low share (below 10^{th} percentile). In practice, we add an interaction term between the realized ex-post default probability and a dummy variable to distinguish between groups. As shown in Fig. B.10, in classes where there was an overall strong and significant correlation, the coefficients are not statistically different between groups of firms with large vs small loans. In other words, the size of the loan relative to the firm's balance sheet does not systematically explain the observed correlation. This is consistent with the assumption that rates contain banks' additional information about firm risk beyond what is captured by public information. For robustness, we repeat the same exercise with the size of total bank loans over total assets (Fig. B.10), to include the possibility that the observed rate on a single loan reflects the ones on all bank loans.

Finally, we want to test that the correlation between rates and the evolution of default probability is stronger in contexts when information is more valuable; in this case, we expect the information component to play a stronger role. Relying on the literature, we use firms' age to distinguish firms for which we expect the unobserved creditworthiness component contained in rates to be very large and dispersed (proxy for risk/growth prospects Haltiwanger, Jarmin,Miranda, 2013); length in the credit market Cloyne, Ferreira, Froemel, Surico, 2023). Older firms are, in general, the survivors, with a good overall credit quality; for young firms, instead, ex-post outcomes might be way more uncertain and dispersed, and banks may put more effort in screening among the to acquire more information on their actual riskiness. We expect the correlation between rates and ex-post outcome to be much stronger for relatively young firms in our sample (below 10^{th} age percentile) than relatively old firms (above

Figure B.10: Interactions of ex-post default probability and loan-size over assets (left) and bank debt over assets (right)



The figure compares the correlation coefficients of residual rates and unexpectedchanges in ex-post default probability across percentiles of i) loan-size over assets (left) and ii) size of bank debt over assets (right) for firms with ex-ante low default probability $< 0.62\%$.

90^{th} age percentile). As shown in Fig. B.11, this is indeed what we find.

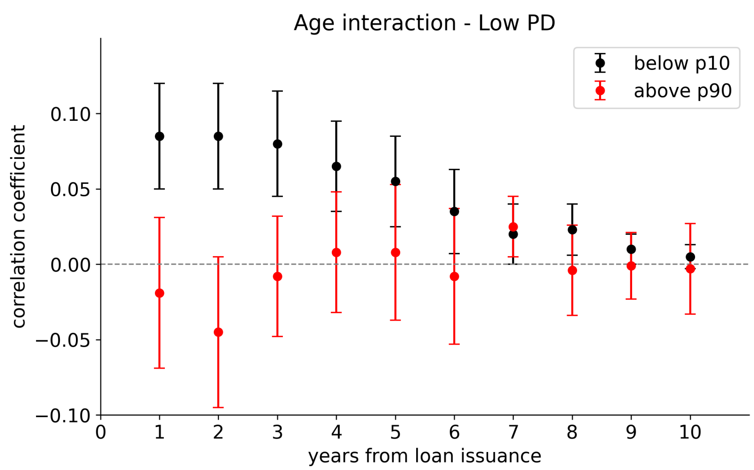


Figure B.11: The figure shows the correlation coefficients of the interactions of ex-post default probability and age (below 10^{th} vs above 90^{th} age percentile). Sample of firms with ex-ante low default probability $< 0.62\%$.

Approach 2: exploiting the difference between fixed and floating rate loans.

We proceed with an exercise to test the transmission of interest rate changes to firm-level risk. We exploit the difference caused by the floating and fixed rate loans on firm's liability, to test whether firms borrowing variable rate loans are more likely to downgrade when reference rates increase. The argument is that the reference rate increases are exogenous to individual firms as they are determined by aggregate monetary policy. The interest rate structure has been used to evaluate the effects of monetary policy on firms' liabilities and outcomes; for example, Gürkaynak et al. (2022) use stock market equity prices to show that investors price-in policy rate changes more when firms have a higher share of floating rate liabilities; Core et al. (2025) also exploit the difference between floating and fixed rate loans to show that the effect of a monetary policy tightening on inflation is weakened if firms facing higher borrowing costs adjust prices. Here, we use a similar strategy to test to which extent interest rates can affect firm's riskiness.

We evaluate the impact of an increase in reference rates following loan origination on firms' creditworthiness after 3 years, using again rating classes transitions and historical default frequencies to approximate the relative increase in default risk. We estimate several specification, including variable rate dummies, yearly increases in reference rates and their interactions with the share of the loan over assets. Unfortunately, we cannot perfectly observe the share of variable rate loans for each firm, so we use the share of variable rate loans in the sample as a proxy. We also use monetary policy shocks as instruments. The baseline specification is the following:

$$\Delta PD_{f,t+3} = \sum_{k=0}^{3} \beta_{1,k} \Delta RefRate_{t+k} + \sum_{k=0}^{3} \beta_{2,k} (\Delta RefRate_{t+k} \times \text{Share Loan}_t) + \gamma_{b,t,p,s} + \epsilon_{f,t} \tag{9}$$

We estimate this specification across sub-samples of loans with maturities exceeding one and two years, to avoid results driven by refinancing. We find no significant impact of increased debt-servicing costs on rating worsening, suggesting that, at least in our sample, an increase in rates inducing an increase in leverage cannot alone induce a worsening firm-level risk. Interestingly, while the rate increase does not induce a worse ex-post outcome, the loan interest rate remains a significant predictor of ex-post riskiness.

However, the validity of our estimates rests on the assumptions that firms do not sort into contract types based on private information or rate expectations (*self-selection*). We mitigates this concern showing that the contract choice is predominantly determined by supply-side lender preferences rather than firm-specific unobservables. Focusing on firms with a rating at the time of the loan, we estimate a linear probability model of the form:

$$Floating_{loan} = \beta X^{observables} + \delta_{q,b,t,s} + \varepsilon_{loan} \tag{10}$$

where $Floating_{loan}$ is a dummy for variable-rate loans and $\delta_{t,b,p,s}$ is the fixed effects interaction term for month, bank, loan purpose and firm sector. Following a variance decomposition approach, we gradually saturate the model with fixed effects. As shown in Table reftab:variance, the lender-quarter dimension alone accounts for approximately 60% of the variation, with the full specification explaining up to 91% of the total variance. This suggests that the contract choice is predominantly driven by supply-side and sectoral factors rather than idiosyncratic firm risk; in fact, banks may prefer to offer floating rate contracts if a variable rate schedule is applied to their own liabilities (see Kirti (2020)).

Table B.3: Variance Decomposition of Variable Rate Choice

Fixed Effects Included	**Adjusted** R^2
Quarter	0.04
+ Lender	0.63
+ Loan Type	0.86
+ Firm Sector	0.91

We also check the balance of observable characteristics between firms taking fixed and variable rate loans. While we find that the two groups differ in some characteristics (variable: fewer firms, larger size, older, riskier, larger financial debt), we include all those characteristics as controls in the following regression, to mitigate the issue of selection bias. We also check the parallel trends assumption by plotting the evolution of rating transitions for firms taking fixed vs variable rate loans, and by including leads of the treatment variable in the regression. We find that rating transitions for firms taking variable rate loans do not differ significantly from those taking fixed rate loans during the period considered. This evidence overall mitigates concerns about selection bias.

Expectations and self selection: firms taking a loan in 2021. To further mitigate the issue of self-selection of firms into fixed or floating rate contracts, we run the exercise at the end of the ZLB period, when firms did not expect a rate hike. We focus on loans issued in 2021, when the policy rate was still at the ZLB and the market did not expect a rate hike. Even for this sample of loans, we do not find a significant effect of reference rate increases on rating worsening. This evidence further supports the idea that the observed correlation is not driven by a mechanical effect of rates on firm fundamentals, but rather by banks' information acquisition.